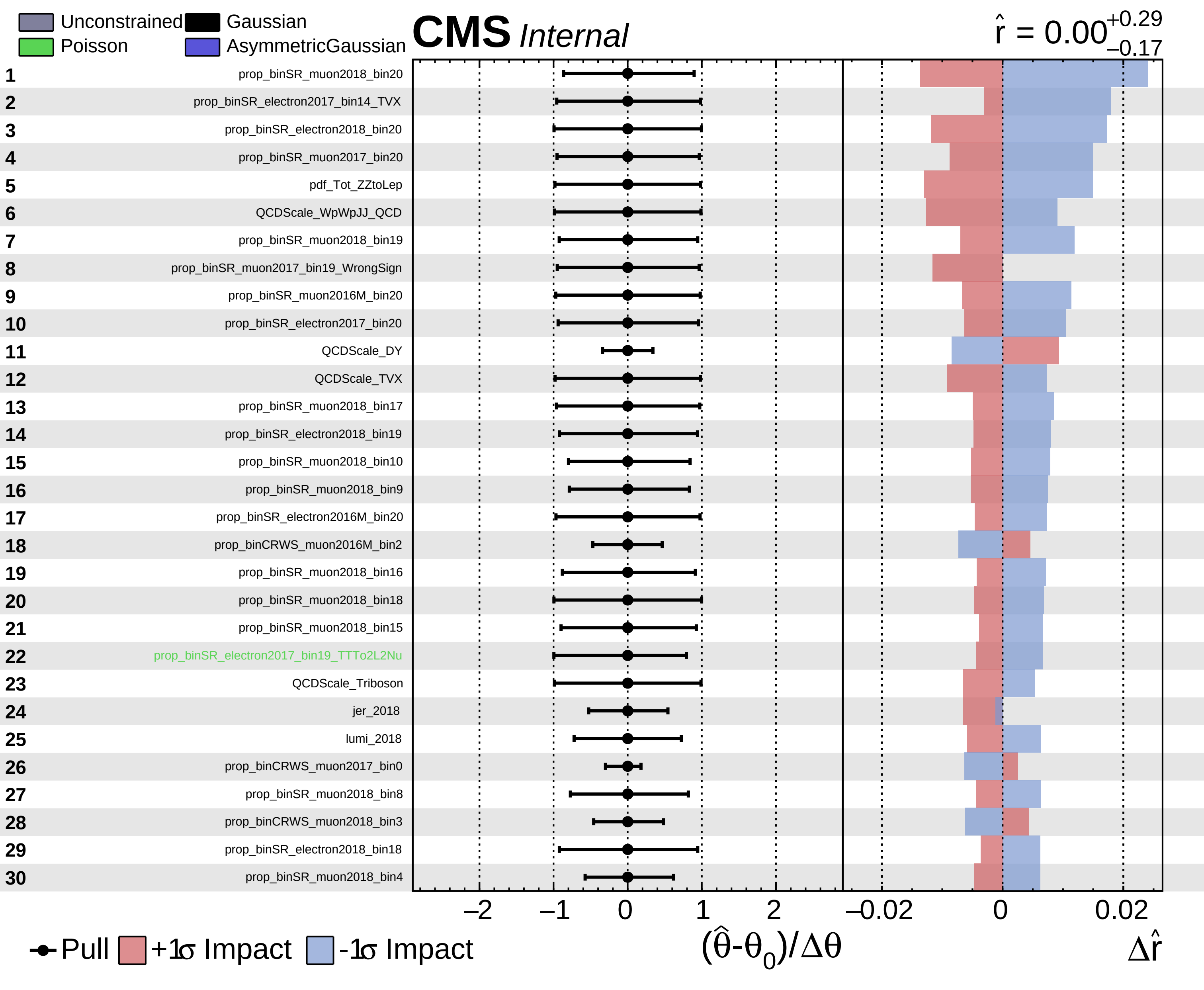
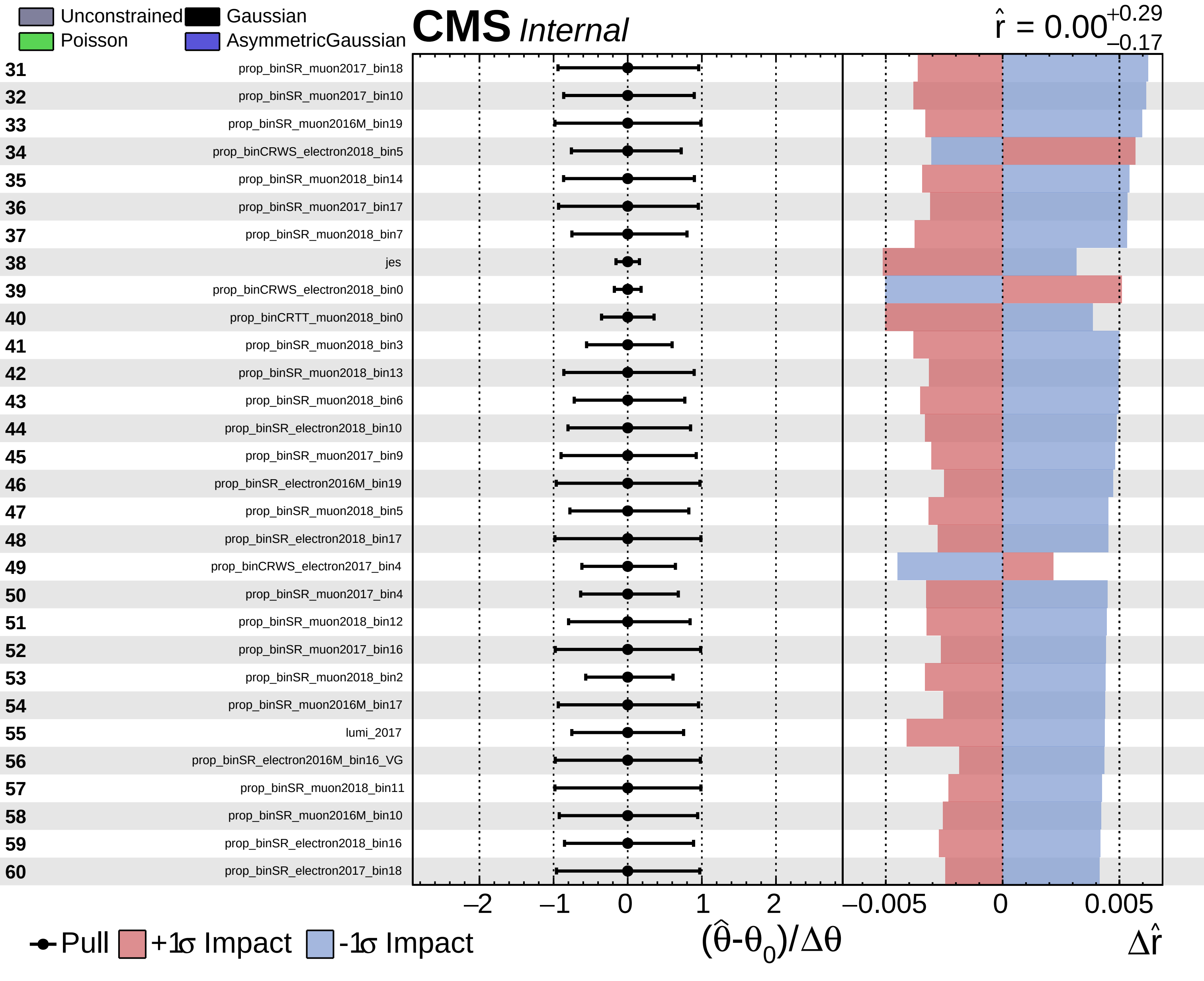
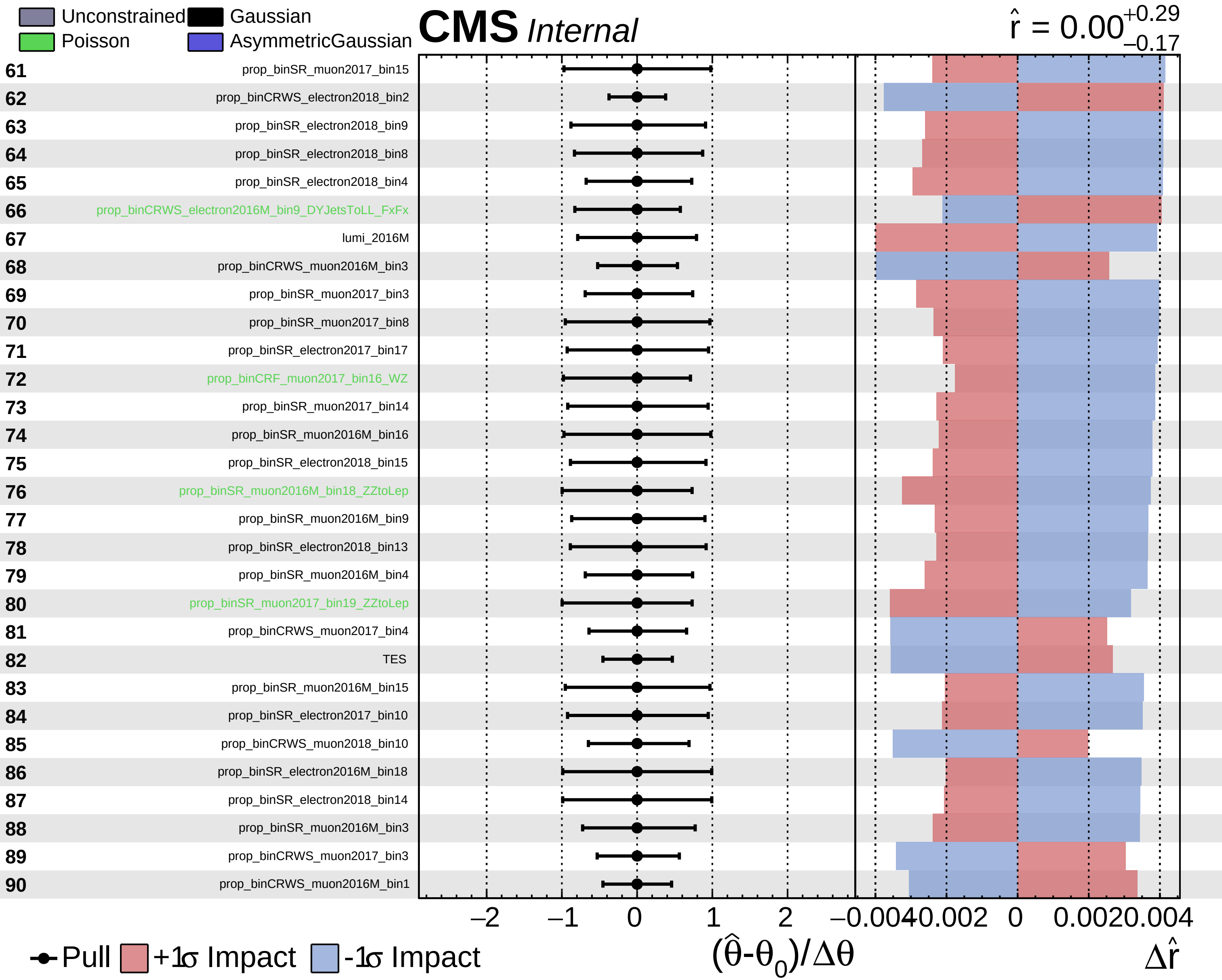
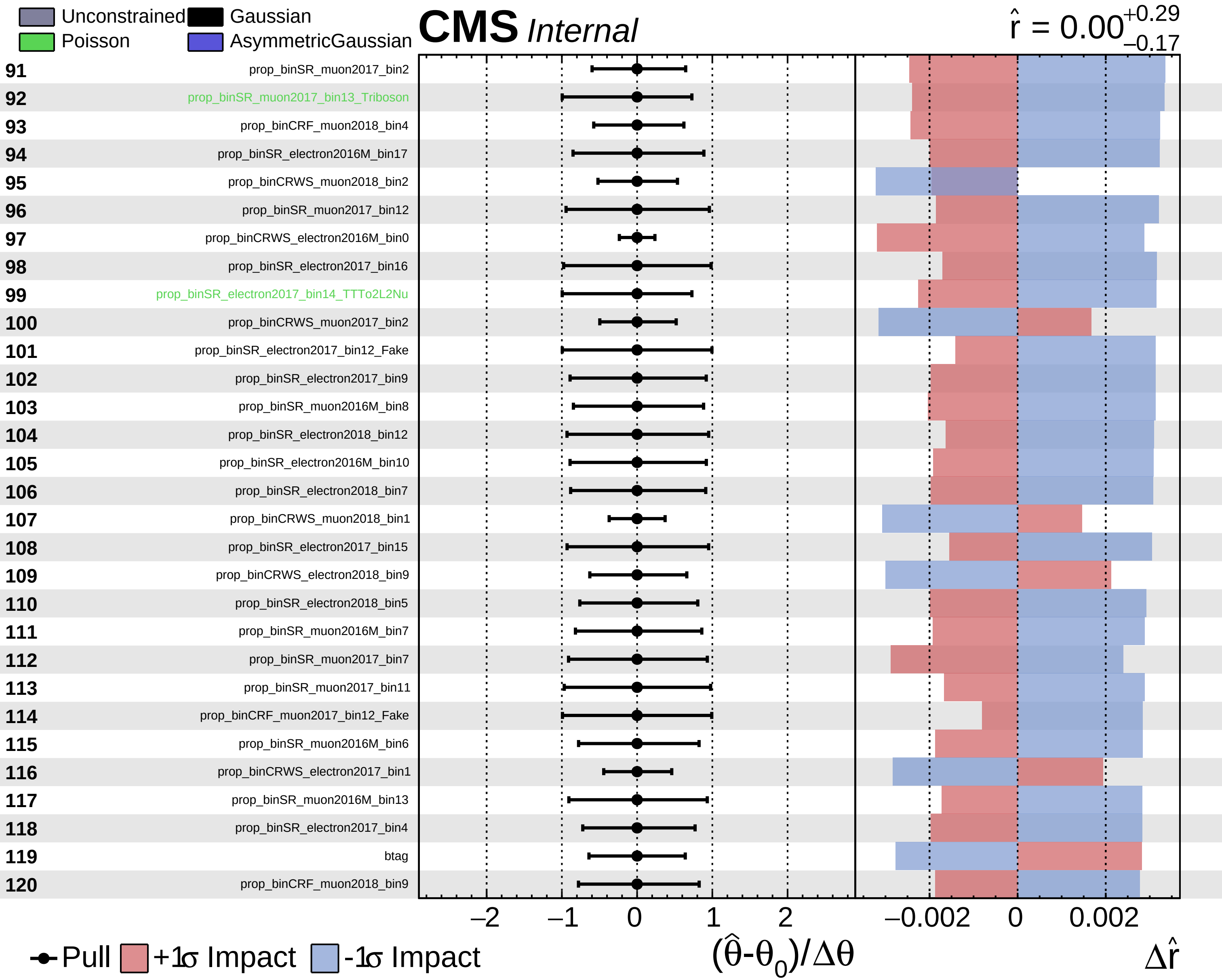


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
1 prop_binSR_muon2018_bin20
2 prop_binSR_electron2017_bin14_TVX
3 prop_binSR_electron2018_bin20
4 prop_binSR_muon2017_bin20
5 pdf_Tot_ZZtoLep
6 QCDScale_WpWpJJ_QCD
7 prop_binSR_muon2018_bin19
8 prop_binSR_muon2017_bin19_WrongSign
9 prop_binSR_muon2016M_bin20
10 prop_binSR_electron2017_bin20
11 QCDScale_DY
12 QCDScale_TVX
13 prop_binSR_muon2018_bin17
14 prop_binSR_electron2018_bin19
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2018_bin9
17 prop_binSR_electron2016M_bin20
18 prop_binCRWS_muon2016M_bin2
19 prop_binSR_muon2018_bin16
20 prop_binSR_muon2018_bin18
21 prop_binSR_muon2018_bin15
22 prop_binSR_electron2017_bin19_TTTo2L2Nu
23 QCDScale_Triboson
24 jer_2018
25 lumi_2018
26 prop_binCRWS_muon2017_bin0
27 prop_binSR_muon2018_bin8
28 prop_binCRWS_muon2018_bin3
29 prop_binSR_electron2018_bin18
30 prop_binSR_muon2018_bin4
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

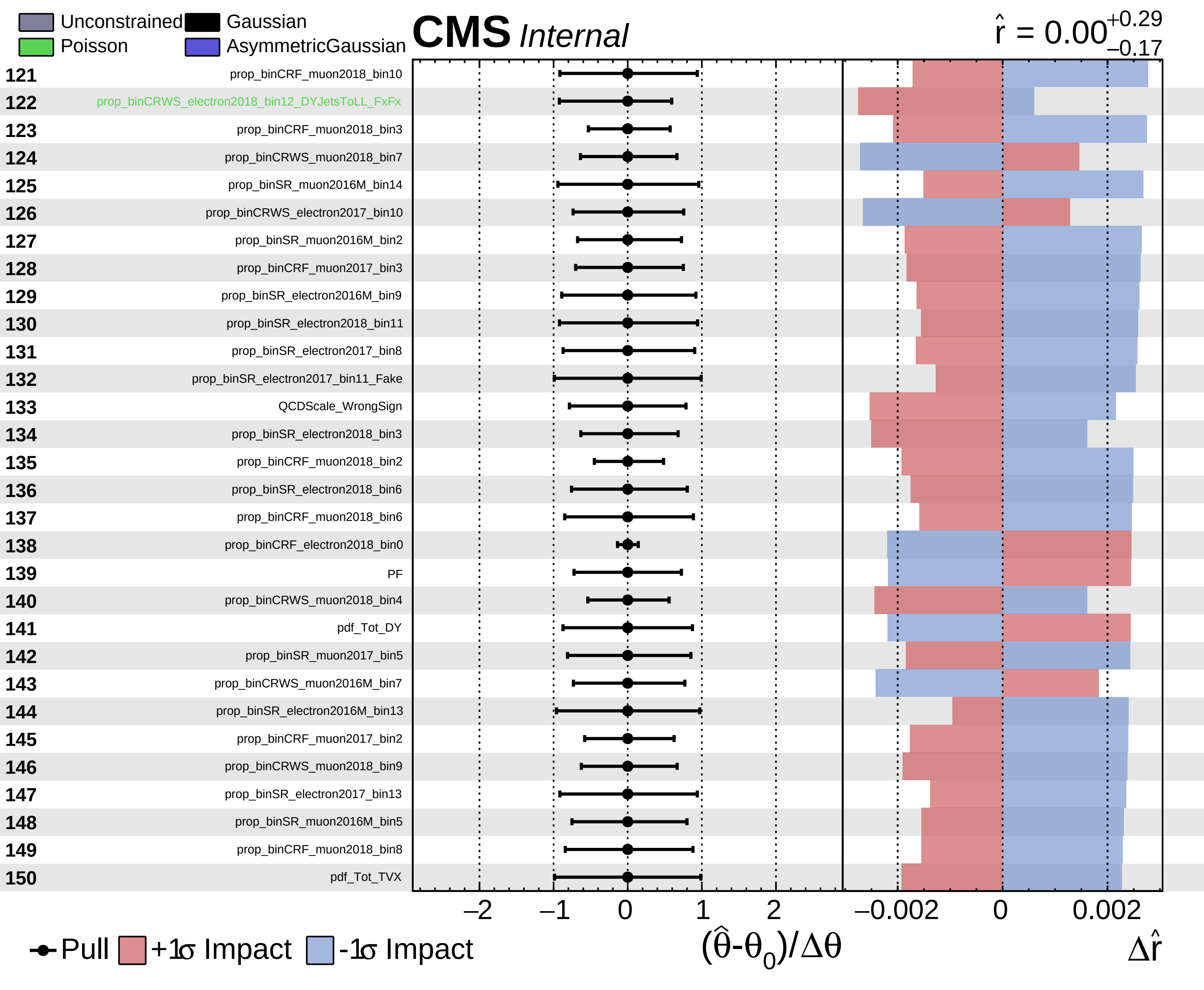


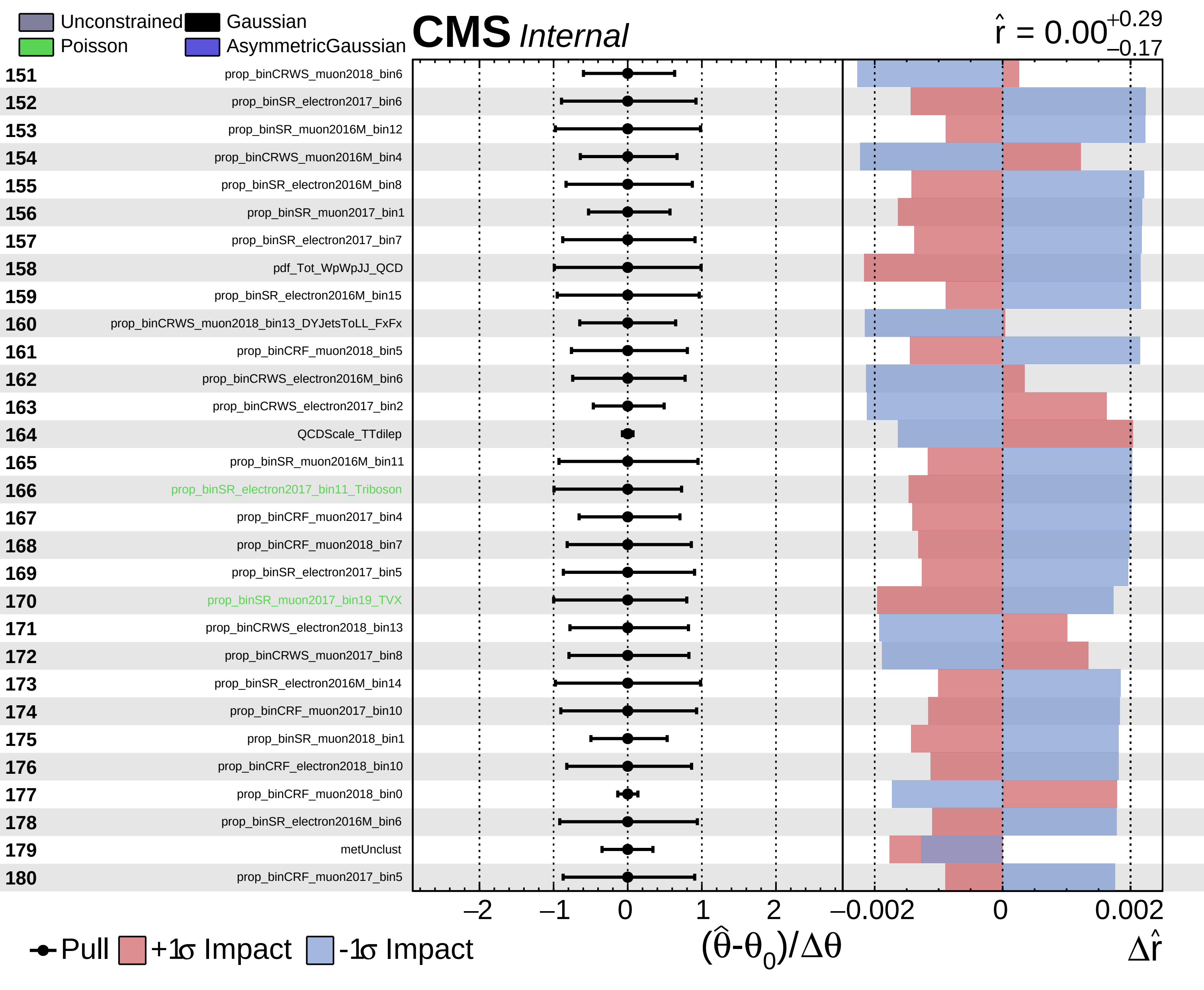
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
31 prop_binSR_muon2017_bin18
32 prop_binSR_muon2017_bin10
33 prop_binSR_muon2016M_bin19
34 prop_binCRWS_electron2018_bin5
35 prop_binSR_muon2018_bin14
36 prop_binSR_muon2017_bin17
37 prop_binSR_muon2018_bin7
38 jes
39 prop_binCRWS_electron2018_bin0
40 prop_binCRTT_muon2018_bin0
41 prop_binSR_muon2018_bin3
42 prop_binSR_muon2018_bin13
43 prop_binSR_muon2018_bin6
44 prop_binSR_electron2018_bin10
45 prop_binSR_muon2017_bin9
46 prop_binSR_electron2016M_bin19
47 prop_binSR_muon2018_bin5
48 prop_binSR_electron2018_bin17
49 prop_binCRWS_electron2017_bin4
50 prop_binSR_muon2017_bin4
51 prop_binSR_muon2018_bin12
52 prop_binSR_muon2017_bin16
53 prop_binSR_muon2018_bin2
54 prop_binSR_muon2016M_bin17
55 lumi_2017
56 prop_binSR_electron2016M_bin16_VG
57 prop_binSR_muon2018_bin11
58 prop_binSR_muon2016M_bin10
59 prop_binSR_electron2018_bin16
60 prop_binSR_electron2017_bin18
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

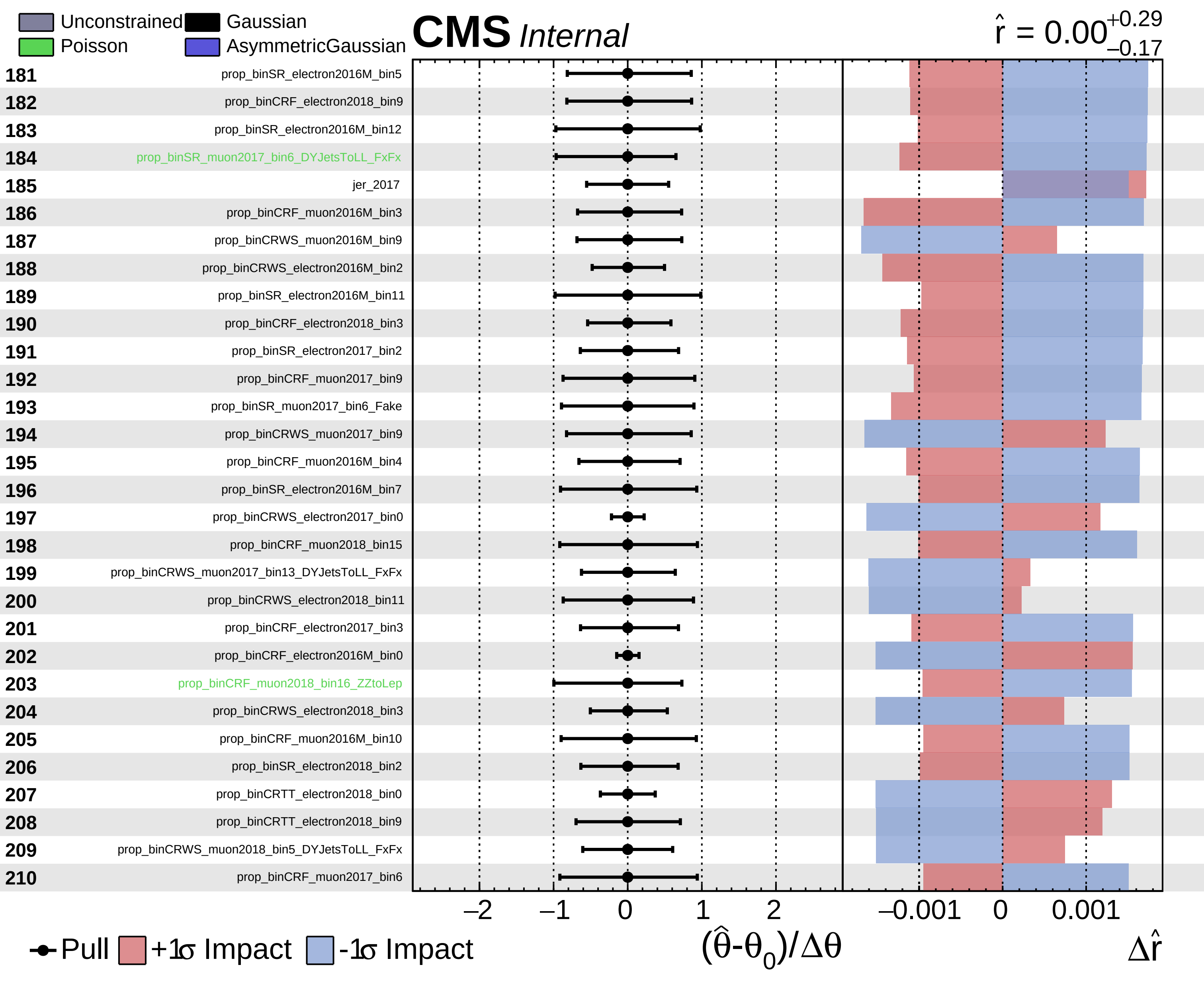

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
61 prop_binSR_muon2017_bin15
62 prop_binCRWS_electron2018_bin2
63 prop_binSR_electron2018_bin9
64 prop_binSR_electron2018_bin8
65 prop_binSR_electron2018_bin4
66 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
67 lumi_2016M
68 prop_binCRWS_muon2016M_bin3
69 prop_binSR_muon2017_bin3
70 prop_binSR_muon2017_bin8
71 prop_binSR_electron2017_bin17
72 prop_binCRF_muon2017_bin16_WZ
73 prop_binSR_muon2017_bin14
74 prop_binSR_muon2016M_bin16
75 prop_binSR_electron2018_bin15
76 prop_binSR_muon2016M_bin18_ZZtoLep
77 prop_binSR_muon2016M_bin9
78 prop_binSR_electron2018_bin13
79 prop_binSR_muon2016M_bin4
80 prop_binSR_muon2017_bin19_ZZtoLep
81 prop_binCRWS_muon2017_bin4
82 TES
83 prop_binSR_muon2016M_bin15
84 prop_binSR_electron2017_bin10
85 prop_binCRWS_muon2018_bin10
86 prop_binSR_electron2016M_bin18
87 prop_binSR_electron2018_bin14
88 prop_binSR_muon2016M_bin3
89 prop_binCRWS_muon2017_bin3
90 prop_binCRWS_muon2016M_bin1
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

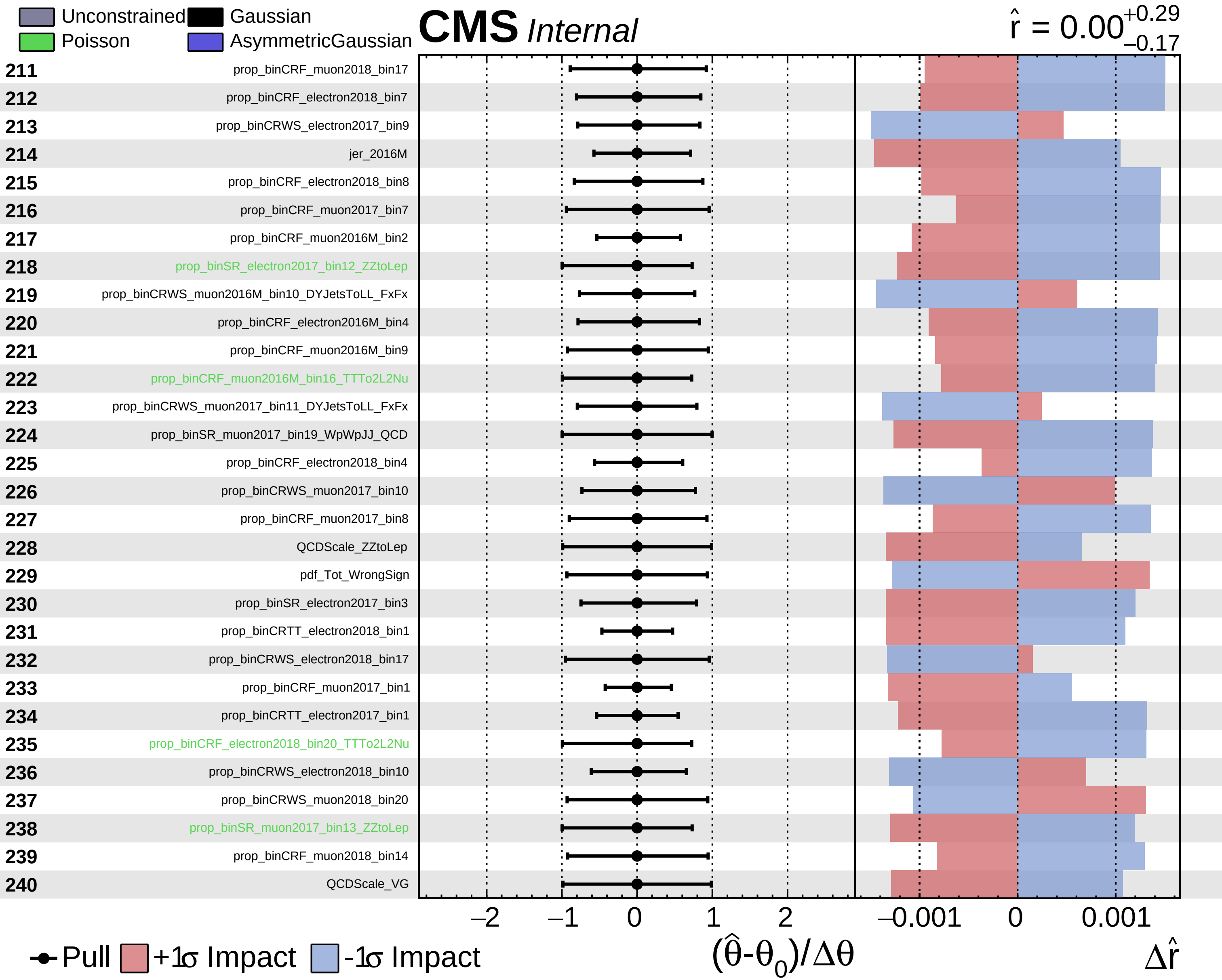


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
91 prop_binSR_muon2017_bin2
92 prop_binSR_muon2017_bin13_Triboson
93 prop_binCRF_muon2018_bin4
94 prop_binSR_electron2016M_bin17
95 prop_binCRWS_muon2018_bin2
96 prop_binSR_muon2017_bin12
97 prop_binCRWS_electron2016M_bin0
98 prop_binSR_electron2017_bin16
99 prop_binSR_electron2017_bin14_TTTo2L2Nu
100 prop_binCRWS_muon2017_bin2
101 prop_binSR_electron2017_bin12_Fake
102 prop_binSR_electron2017_bin9
103 prop_binSR_muon2016M_bin8
104 prop_binSR_electron2018_bin12
105 prop_binSR_electron2016M_bin10
106 prop_binSR_electron2018_bin7
107 prop_binCRWS_muon2018_bin1
108 prop_binSR_electron2017_bin15
109 prop_binCRWS_electron2018_bin9
110 prop_binSR_electron2018_bin5
111 prop_binSR_muon2016M_bin7
112 prop_binSR_muon2017_bin7
113 prop_binSR_muon2017_bin11
114 prop_binCRF_muon2017_bin12_Fake
115 prop_binSR_muon2016M_bin6
116 prop_binCRWS_electron2017_bin1
117 prop_binSR_muon2016M_bin13
118 prop_binSR_electron2017_bin4
119 btag
120 prop_binCRF_muon2018_bin9
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

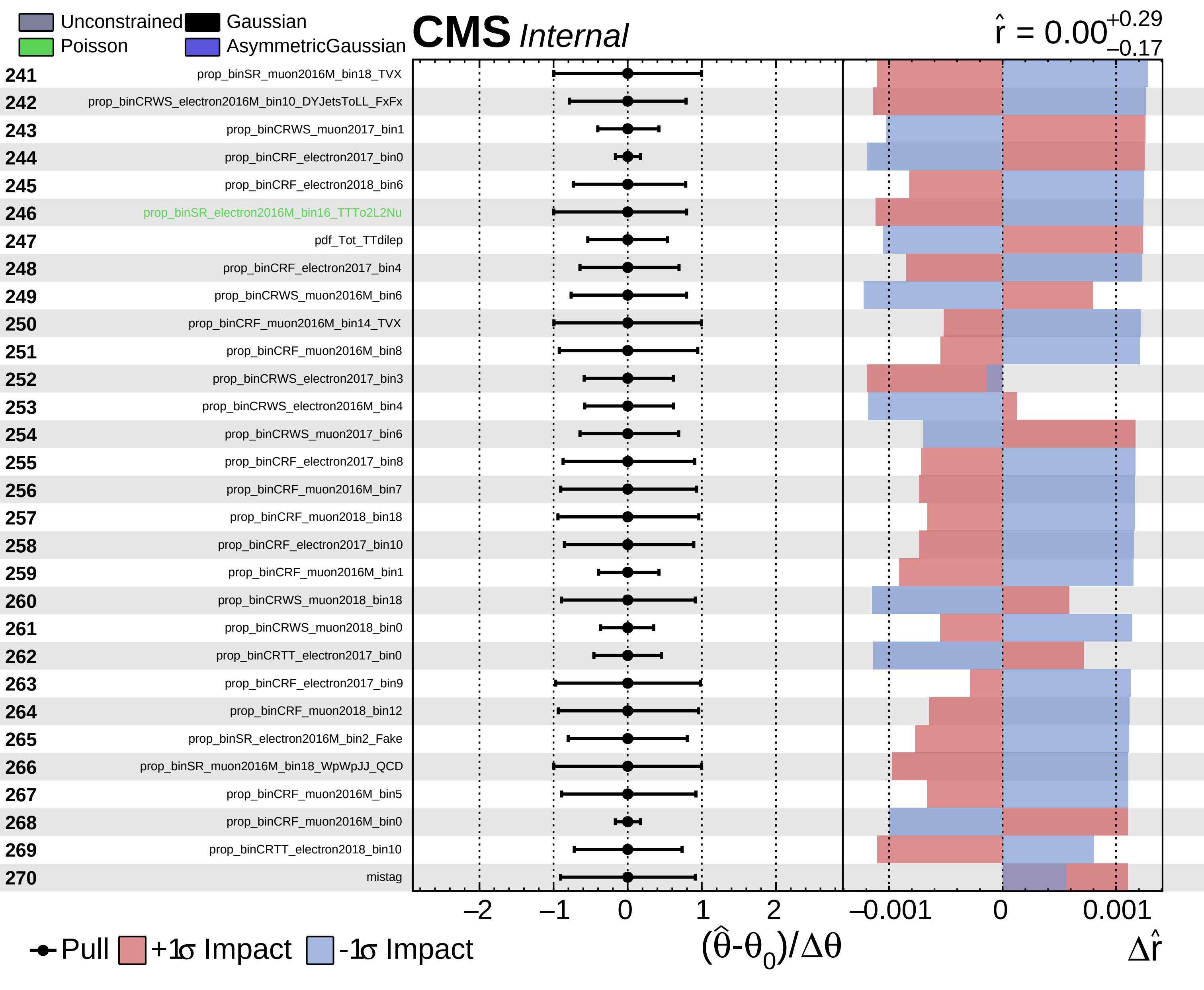

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
121 prop_binCRF_muon2018_bin10
122 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
123 prop_binCRF_muon2018_bin3
124 prop_binCRWS_muon2018_bin7
125 prop_binSR_muon2016M_bin14
126 prop_binCRWS_electron2017_bin10
127 prop_binSR_muon2016M_bin2
128 prop_binCRF_muon2017_bin3
129 prop_binSR_electron2016M_bin9
130 prop_binSR_electron2018_bin11
131 prop_binSR_electron2017_bin8
132 prop_binSR_electron2017_bin11_Fake
133 QCDScale_WrongSign
134 prop_binSR_electron2018_bin3
135 prop_binCRF_muon2018_bin2
136 prop_binSR_electron2018_bin6
137 prop_binCRF_muon2018_bin6
138 prop_binCRF_electron2018_bin0
139 PF
140 prop_binCRWS_muon2018_bin4
141 pdf_Tot_DY
142 prop_binSR_muon2017_bin5
143 prop_binCRWS_muon2016M_bin7
144 prop_binSR_electron2016M_bin13
145 prop_binCRF_muon2017_bin2
146 prop_binCRWS_muon2018_bin9
147 prop_binSR_electron2017_bin13
148 prop_binSR_muon2016M_bin5
149 prop_binCRF_muon2018_bin8
150 pdf_Tot_TVX
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
151 prop_binCRWS_muon2018_bin6
152 prop_binSR_electron2017_bin6
153 prop_binSR_muon2016M_bin12
154 prop_binCRWS_muon2016M_bin4
155 prop_binSR_electron2016M_bin8
156 prop_binSR_muon2017_bin1
157 prop_binSR_electron2017_bin7
158 pdf_Tot_WpWpJJ_QCD
159 prop_binSR_electron2016M_bin15
160 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
161 prop_binCRF_muon2018_bin5
162 prop_binCRWS_electron2016M_bin6
163 prop_binCRWS_electron2017_bin2
164 QCDScale_TTdilep
165 prop_binSR_muon2016M_bin11
166 prop_binSR_electron2017_bin11_Triboson
167 prop_binCRF_muon2017_bin4
168 prop_binCRF_muon2018_bin7
169 prop_binSR_electron2017_bin5
170 prop_binSR_muon2017_bin19_TVX
171 prop_binCRWS_electron2018_bin13
172 prop_binCRWS_muon2017_bin8
173 prop_binSR_electron2016M_bin14
174 prop_binCRF_muon2017_bin10
175 prop_binSR_muon2018_bin1
176 prop_binCRF_electron2018_bin10
177 prop_binCRF_muon2018_bin0
178 prop_binSR_electron2016M_bin6
179 metUnclust
180 prop_binCRF_muon2017_bin5
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

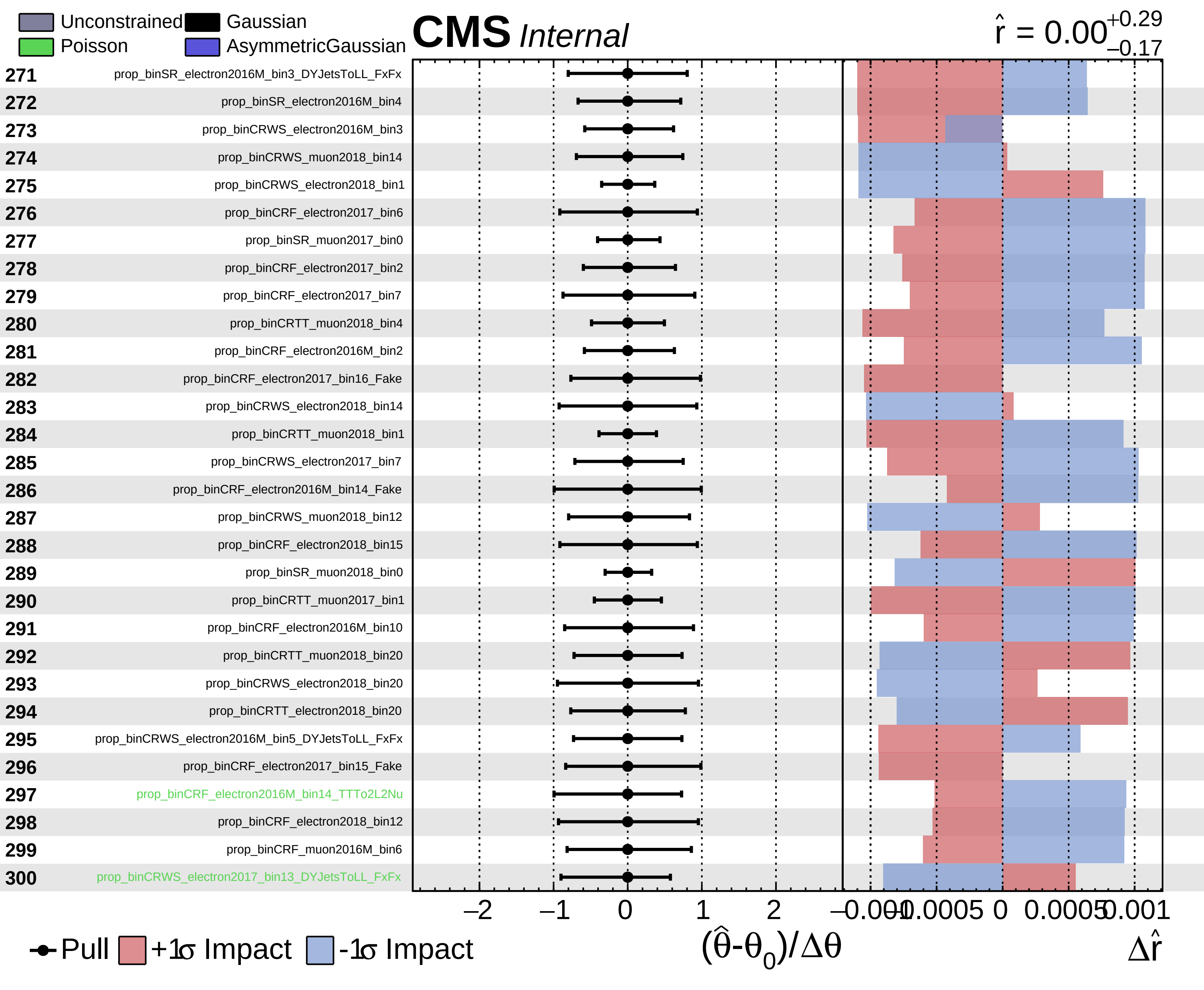

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
181 prop_binSR_electron2016M_bin5
182 prop_binCRF_electron2018_bin9
183 prop_binSR_electron2016M_bin12
184 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
185 jer_2017
186 prop_binCRF_muon2016M_bin3
187 prop_binCRWS_muon2016M_bin9
188 prop_binCRWS_electron2016M_bin2
189 prop_binSR_electron2016M_bin11
190 prop_binCRF_electron2018_bin3
191 prop_binSR_electron2017_bin2
192 prop_binCRF_muon2017_bin9
193 prop_binSR_muon2017_bin6_Fake
194 prop_binCRWS_muon2017_bin9
195 prop_binCRF_muon2016M_bin4
196 prop_binSR_electron2016M_bin7
197 prop_binCRWS_electron2017_bin0
198 prop_binCRF_muon2018_bin15
199 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
200 prop_binCRWS_electron2018_bin11
201 prop_binCRF_electron2017_bin3
202 prop_binCRF_electron2016M_bin0
203 prop_binCRF_muon2018_bin16_ZZtoLep
204 prop_binCRWS_electron2018_bin3
205 prop_binCRF_muon2016M_bin10
206 prop_binSR_electron2018_bin2
207 prop_binCRTT_electron2018_bin0
208 prop_binCRTT_electron2018_bin9
209 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
210 prop_binCRF_muon2017_bin6
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

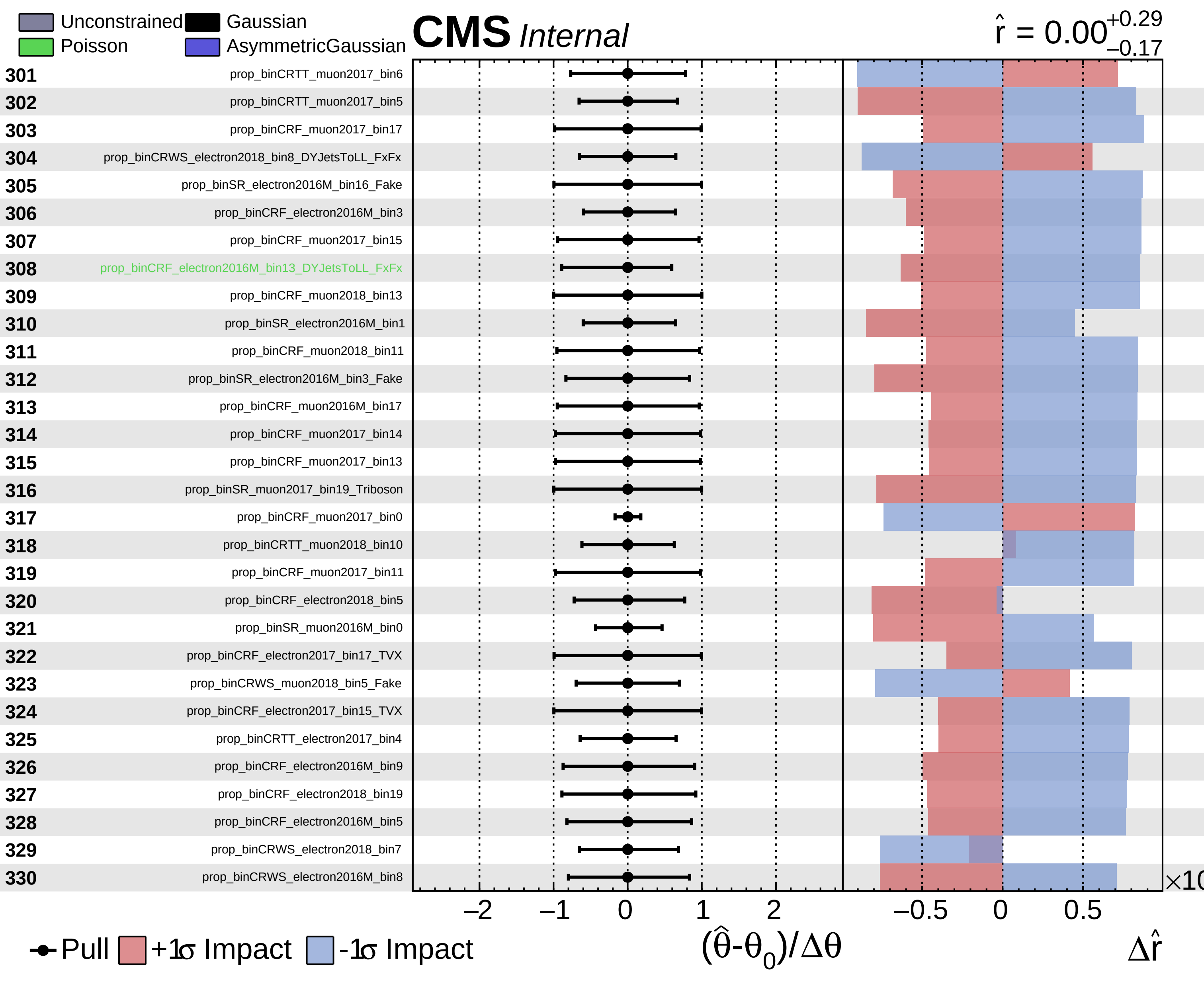


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
211 prop_binCRF_muon2018_bin17
212 prop_binCRF_electron2018_bin7
213 prop_binCRWS_electron2017_bin9
214 jer_2016M
215 prop_binCRF_electron2018_bin8
216 prop_binCRF_muon2017_bin7
217 prop_binCRF_muon2016M_bin2
218 prop_binSR_electron2017_bin12_ZZtoLep
219 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
220 prop_binCRF_electron2016M_bin4
221 prop_binCRF_muon2016M_bin9
222 prop_binCRF_muon2016M_bin16_TTTo2L2Nu
223 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
224 prop_binSR_muon2017_bin19_WpWpJJ_QCD
225 prop_binCRF_electron2018_bin4
226 prop_binCRWS_muon2017_bin10
227 prop_binCRF_muon2017_bin8
228 QCDScale_ZZtoLep
229 pdf_Tot_WrongSign
230 prop_binSR_electron2017_bin3
231 prop_binCRTT_electron2018_bin1
232 prop_binCRWS_electron2018_bin17
233 prop_binCRF_muon2017_bin1
234 prop_binCRTT_electron2017_bin1
235 prop_binCRF_electron2018_bin20_TTTo2L2Nu
236 prop_binCRWS_electron2018_bin10
237 prop_binCRWS_muon2018_bin20
238 prop_binSR_muon2017_bin13_ZZtoLep
239 prop_binCRF_muon2018_bin14
240 QCDScale_VG
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

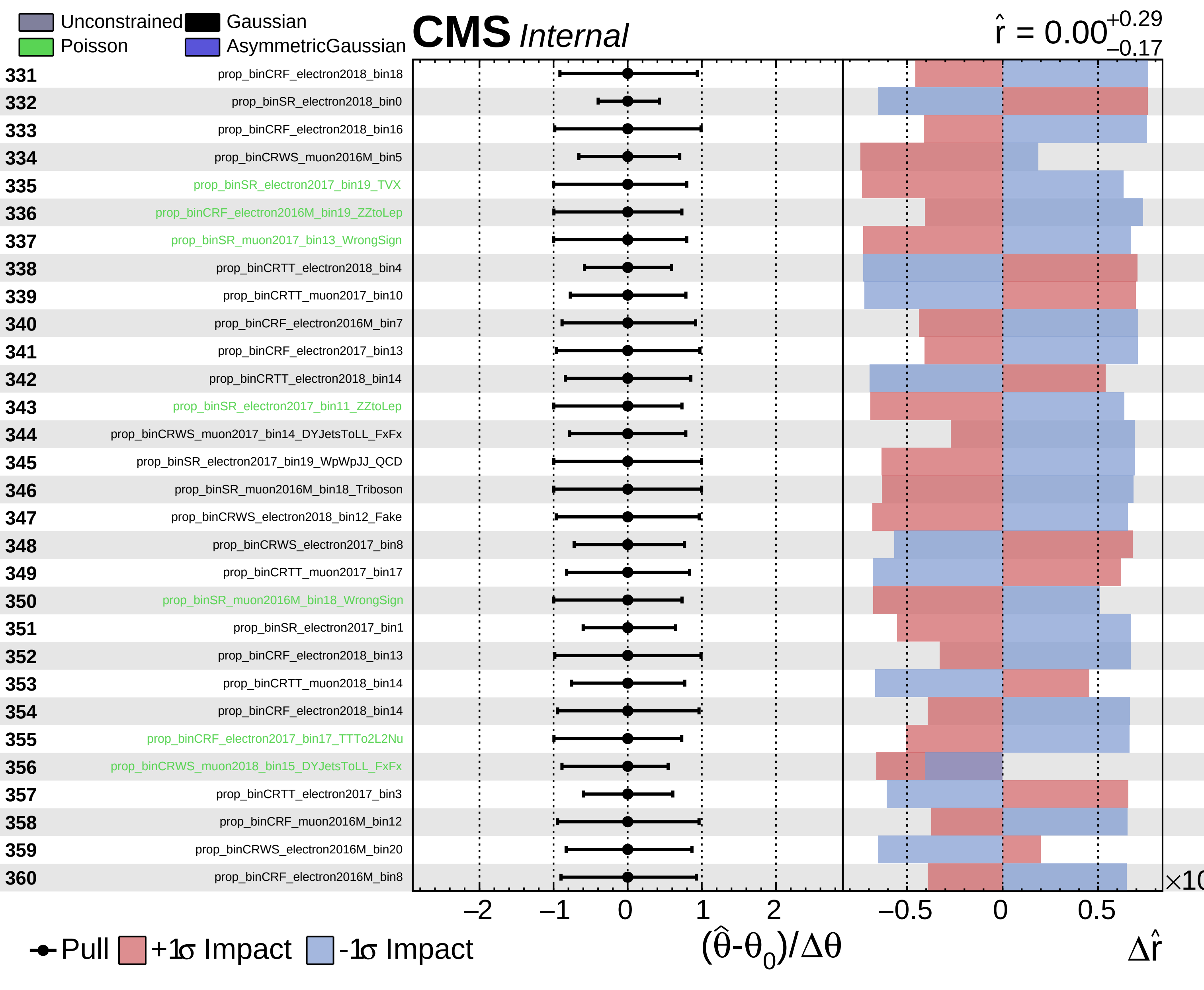


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
241 prop_binSR_muon2016M_bin18_TVX
242 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
243 prop_binCRWS_muon2017_bin1
244 prop_binCRF_electron2017_bin0
245 prop_binCRF_electron2018_bin6
246 prop_binSR_electron2016M_bin16_TTTo2L2Nu
247 pdf_Tot_TTdilep
248 prop_binCRF_electron2017_bin4
249 prop_binCRWS_muon2016M_bin6
250 prop_binCRF_muon2016M_bin14_TVX
251 prop_binCRF_muon2016M_bin8
252 prop_binCRWS_electron2017_bin3
253 prop_binCRWS_electron2016M_bin4
254 prop_binCRWS_muon2017_bin6
255 prop_binCRF_electron2017_bin8
256 prop_binCRF_muon2016M_bin7
257 prop_binCRF_muon2018_bin18
258 prop_binCRF_electron2017_bin10
259 prop_binCRF_muon2016M_bin1
260 prop_binCRWS_muon2018_bin18
261 prop_binCRWS_muon2018_bin0
262 prop_binCRTT_electron2017_bin0
263 prop_binCRF_electron2017_bin9
264 prop_binCRF_muon2018_bin12
265 prop_binSR_electron2016M_bin2_Fake
266 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
267 prop_binCRF_muon2016M_bin5
268 prop_binCRF_muon2016M_bin0
269 prop_binCRTT_electron2018_bin10
270 mistag
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

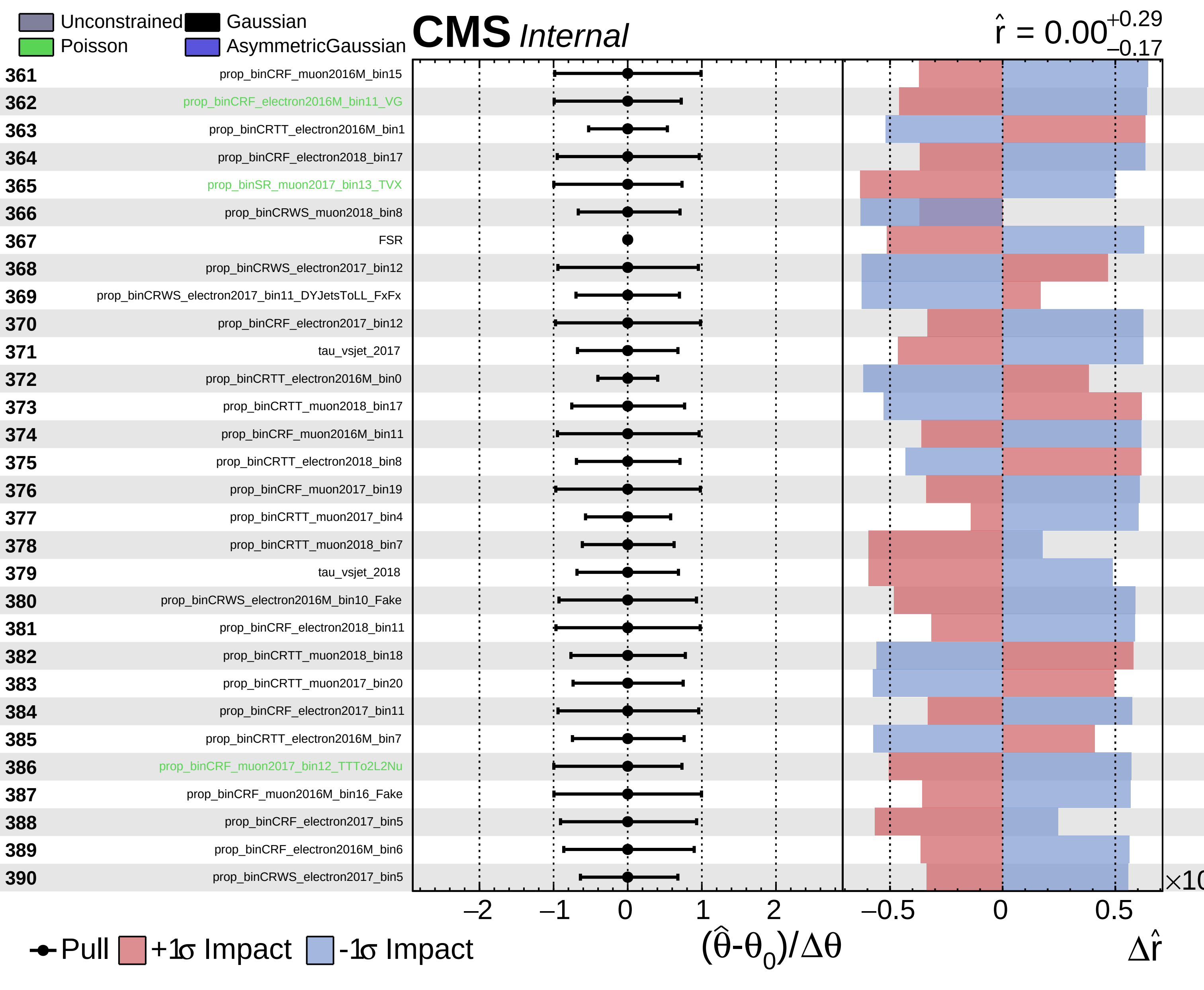

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
271 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
272 prop_binSR_electron2016M_bin4
273 prop_binCRWS_electron2016M_bin3
274 prop_binCRWS_muon2018_bin14
275 prop_binCRWS_electron2018_bin1
276 prop_binCRF_electron2017_bin6
277 prop_binSR_muon2017_bin0
278 prop_binCRF_electron2017_bin2
279 prop_binCRF_electron2017_bin7
280 prop_binCRTT_muon2018_bin4
281 prop_binCRF_electron2016M_bin2
282 prop_binCRF_electron2017_bin16_Fake
283 prop_binCRWS_electron2018_bin14
284 prop_binCRTT_muon2018_bin1
285 prop_binCRWS_electron2017_bin7
286 prop_binCRF_electron2016M_bin14_Fake
287 prop_binCRWS_muon2018_bin12
288 prop_binCRF_electron2018_bin15
289 prop_binSR_muon2018_bin0
290 prop_binCRTT_muon2017_bin1
291 prop_binCRF_electron2016M_bin10
292 prop_binCRTT_muon2018_bin20
293 prop_binCRWS_electron2018_bin20
294 prop_binCRTT_electron2018_bin20
295 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
296 prop_binCRF_electron2017_bin15_Fake
297 prop_binCRF_electron2016M_bin14_TTTo2L2Nu
298 prop_binCRF_electron2018_bin12
299 prop_binCRF_muon2016M_bin6
300 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



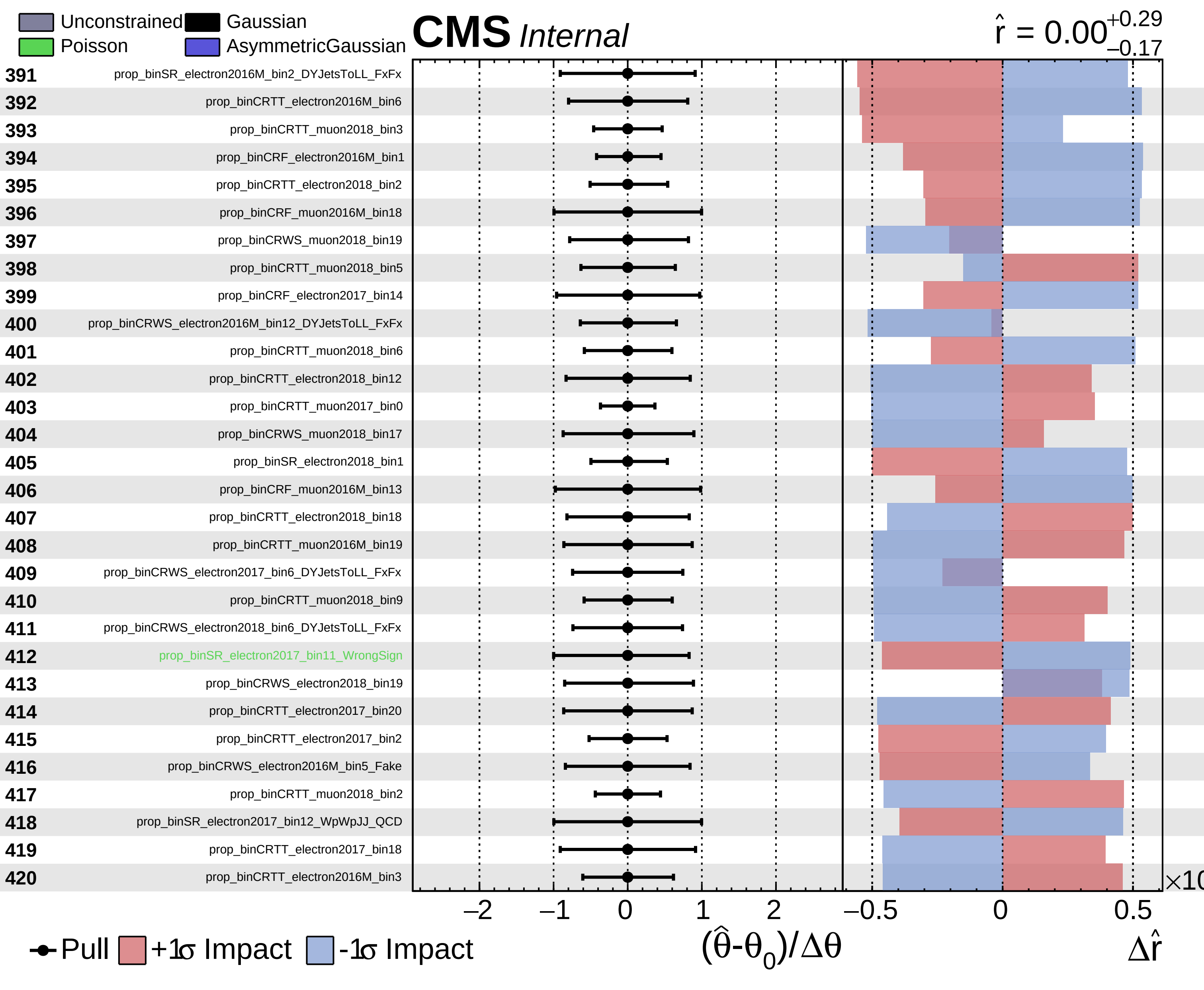
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
301 prop_binCRTT_muon2017_bin6
302 prop_binCRTT_muon2017_bin5
303 prop_binCRF_muon2017_bin17
304 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
305 prop_binSR_electron2016M_bin16_Fake
306 prop_binCRF_electron2016M_bin3
307 prop_binCRF_muon2017_bin15
308 prop_binCRF_electron2016M_bin13_DYJetsToLL_FxFx
309 prop_binCRF_muon2018_bin13
310 prop_binSR_electron2016M_bin1
311 prop_binCRF_muon2018_bin11
312 prop_binSR_electron2016M_bin3_Fake
313 prop_binCRF_muon2016M_bin17
314 prop_binCRF_muon2017_bin14
315 prop_binCRF_muon2017_bin13
316 prop_binSR_muon2017_bin19_Triboson
317 prop_binCRF_muon2017_bin0
318 prop_binCRTT_muon2018_bin10
319 prop_binCRF_muon2017_bin11
320 prop_binCRF_electron2018_bin5
321 prop_binSR_muon2016M_bin0
322 prop_binCRF_electron2017_bin17_TVX
323 prop_binCRWS_muon2018_bin5_Fake
324 prop_binCRF_electron2017_bin15_TVX
325 prop_binCRTT_electron2017_bin4
326 prop_binCRF_electron2016M_bin9
327 prop_binCRF_electron2018_bin19
328 prop_binCRF_electron2016M_bin5
329 prop_binCRWS_electron2018_bin7
330 prop_binCRWS_electron2016M_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



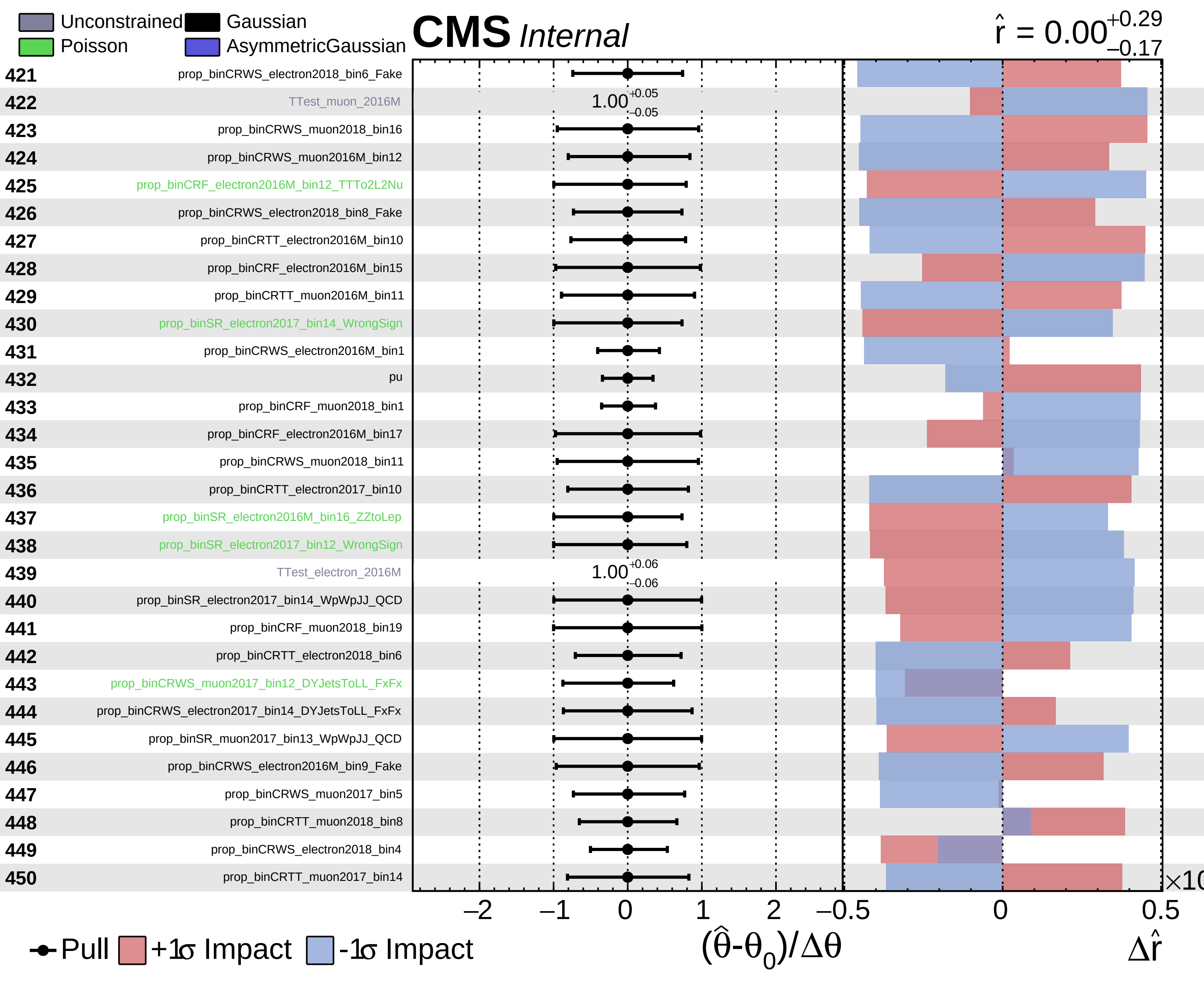
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
331 prop_binCRF_electron2018_bin18
332 prop_binSR_electron2018_bin0
333 prop_binCRF_electron2018_bin16
334 prop_binCRWS_muon2016M_bin5
335 prop_binSR_electron2017_bin19_TVX
336 prop_binCRF_electron2016M_bin19_ZZtoLep
337 prop_binSR_muon2017_bin13_WrongSign
338 prop_binCRTT_electron2018_bin4
339 prop_binCRTT_muon2017_bin10
340 prop_binCRF_electron2016M_bin7
341 prop_binCRF_electron2017_bin13
342 prop_binCRTT_electron2018_bin14
343 prop_binSR_electron2017_bin11_ZZtoLep
344 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
345 prop_binSR_electron2017_bin19_WpWpJJ_QCD
346 prop_binSR_muon2016M_bin18_Triboson
347 prop_binCRWS_electron2018_bin12_Fake
348 prop_binCRWS_electron2017_bin8
349 prop_binCRTT_muon2017_bin17
350 prop_binSR_muon2016M_bin18_WrongSign
351 prop_binSR_electron2017_bin1
352 prop_binCRF_electron2018_bin13
353 prop_binCRTT_muon2018_bin14
354 prop_binCRF_electron2018_bin14
355 prop_binCRF_electron2017_bin17_TTTo2L2Nu
356 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
357 prop_binCRTT_electron2017_bin3
358 prop_binCRF_muon2016M_bin12
359 prop_binCRWS_electron2016M_bin20
360 prop_binCRF_electron2016M_bin8
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



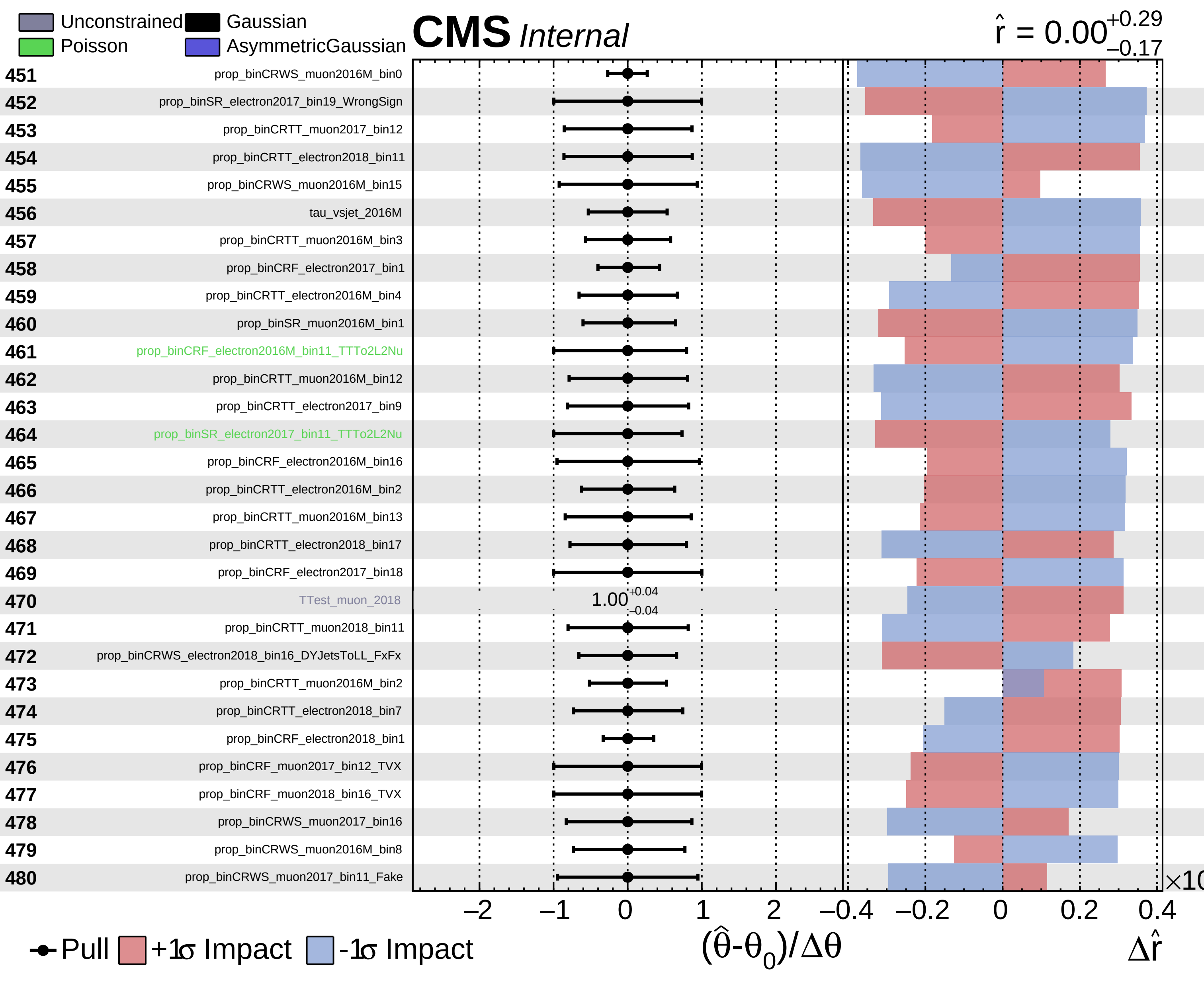
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
361 prop_binCRF_muon2016M_bin15
362 prop_binCRF_electron2016M_bin11_VG
363 prop_binCRTT_electron2016M_bin1
364 prop_binCRF_electron2018_bin17
365 prop_binSR_muon2017_bin13_TVX
366 prop_binCRWS_muon2018_bin8
367 FSR
368 prop_binCRWS_electron2017_bin12
369 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
370 prop_binCRF_electron2017_bin12
371 tau_vsjet_2017
372 prop_binCRTT_electron2016M_bin0
373 prop_binCRTT_muon2018_bin17
374 prop_binCRF_muon2016M_bin11
375 prop_binCRTT_electron2018_bin8
376 prop_binCRF_muon2017_bin19
377 prop_binCRTT_muon2017_bin4
378 prop_binCRTT_muon2018_bin7
379 tau_vsjet_2018
380 prop_binCRWS_electron2016M_bin10_Fake
381 prop_binCRF_electron2018_bin11
382 prop_binCRTT_muon2018_bin18
383 prop_binCRTT_muon2017_bin20
384 prop_binCRF_electron2017_bin11
385 prop_binCRTT_electron2016M_bin7
386 prop_binCRF_muon2017_bin12_TTTo2L2Nu
387 prop_binCRF_muon2016M_bin16_Fake
388 prop_binCRF_electron2017_bin5
389 prop_binCRF_electron2016M_bin6
390 prop_binCRWS_electron2017_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



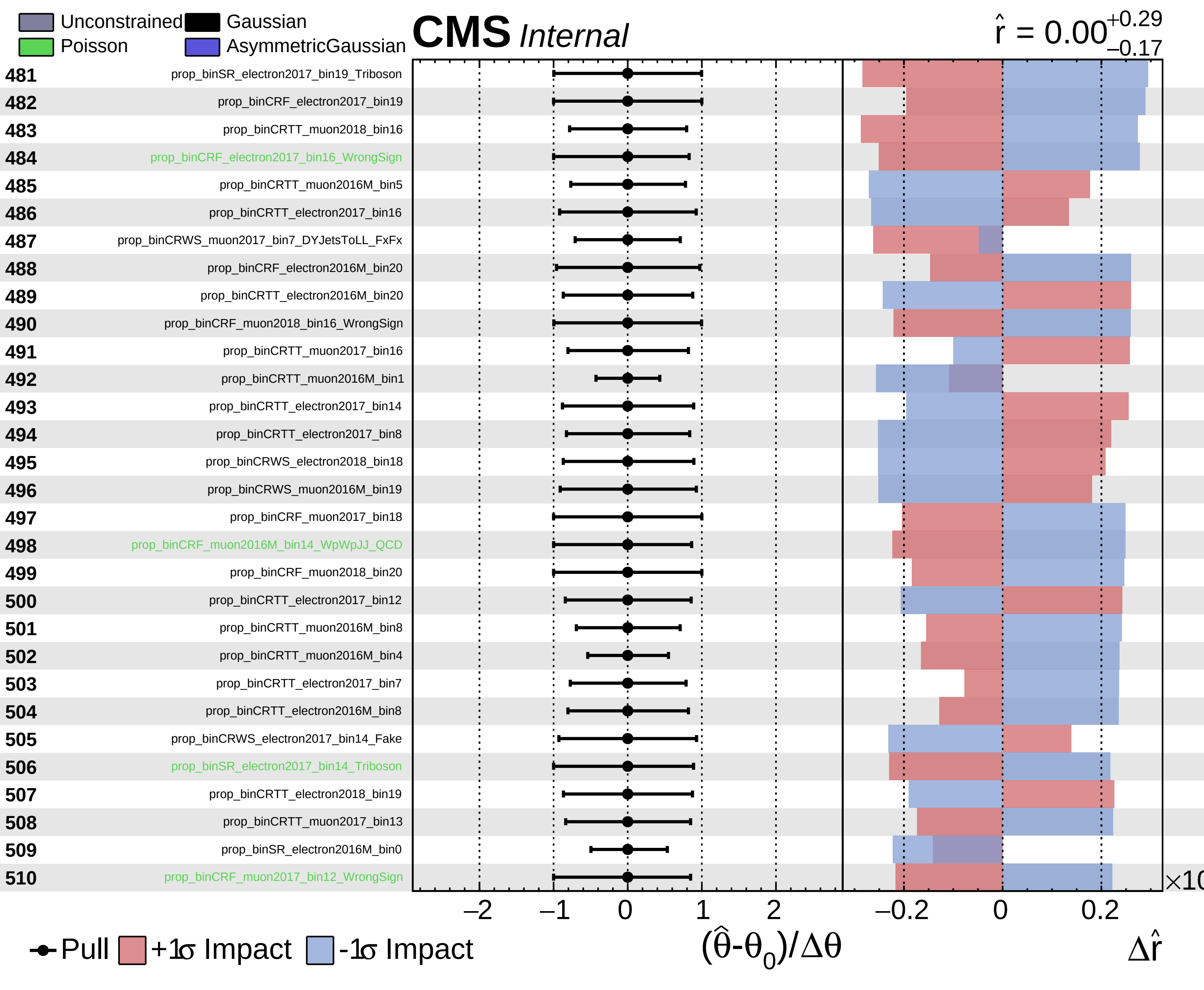
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
391 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
392 prop_binCRTT_electron2016M_bin6
393 prop_binCRTT_muon2018_bin3
394 prop_binCRF_electron2016M_bin1
395 prop_binCRTT_electron2018_bin2
396 prop_binCRF_muon2016M_bin18
397 prop_binCRWS_muon2018_bin19
398 prop_binCRTT_muon2018_bin5
399 prop_binCRF_electron2017_bin14
400 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
401 prop_binCRTT_muon2018_bin6
402 prop_binCRTT_electron2018_bin12
403 prop_binCRTT_muon2017_bin0
404 prop_binCRWS_muon2018_bin17
405 prop_binSR_electron2018_bin1
406 prop_binCRF_muon2016M_bin13
407 prop_binCRTT_electron2018_bin18
408 prop_binCRTT_muon2016M_bin19
409 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
410 prop_binCRTT_muon2018_bin9
411 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
412 prop_binSR_electron2017_bin11_WrongSign
413 prop_binCRWS_electron2018_bin19
414 prop_binCRTT_electron2017_bin20
415 prop_binCRTT_electron2017_bin2
416 prop_binCRWS_electron2016M_bin5_Fake
417 prop_binCRTT_muon2018_bin2
418 prop_binSR_electron2017_bin12_WpWpJJ_QCD
419 prop_binCRTT_electron2017_bin18
420 prop_binCRTT_electron2016M_bin3
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



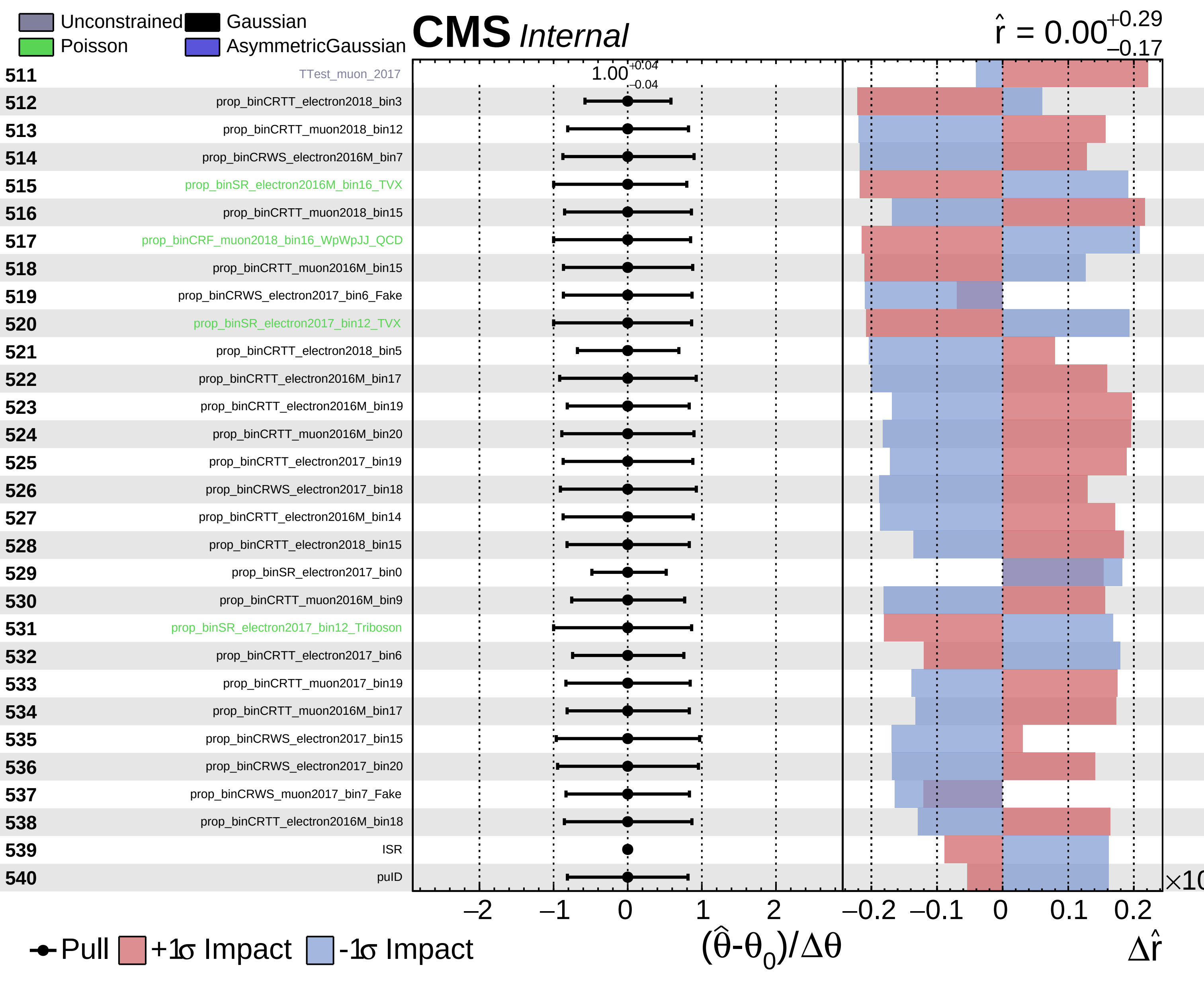
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
421 prop_binCRWS_electron2018_bin6_Fake
422 TTest_muon_2016M
1.00 +0.05 −0.05
423 prop_binCRWS_muon2018_bin16
424 prop_binCRWS_muon2016M_bin12
425 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
426 prop_binCRWS_electron2018_bin8_Fake
427 prop_binCRTT_electron2016M_bin10
428 prop_binCRF_electron2016M_bin15
429 prop_binCRTT_muon2016M_bin11
430 prop_binSR_electron2017_bin14_WrongSign
431 prop_binCRWS_electron2016M_bin1
432 pu
433 prop_binCRF_muon2018_bin1
434 prop_binCRF_electron2016M_bin17
435 prop_binCRWS_muon2018_bin11
436 prop_binCRTT_electron2017_bin10
437 prop_binSR_electron2016M_bin16_ZZtoLep
438 prop_binSR_electron2017_bin12_WrongSign
439 TTest_electron_2016M
1.00 +0.06 −0.06
440 prop_binSR_electron2017_bin14_WpWpJJ_QCD
441 prop_binCRF_muon2018_bin19
442 prop_binCRTT_electron2018_bin6
443 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
444 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
445 prop_binSR_muon2017_bin13_WpWpJJ_QCD
446 prop_binCRWS_electron2016M_bin9_Fake
447 prop_binCRWS_muon2017_bin5
448 prop_binCRTT_muon2018_bin8
449 prop_binCRWS_electron2018_bin4
450 prop_binCRTT_muon2017_bin14
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



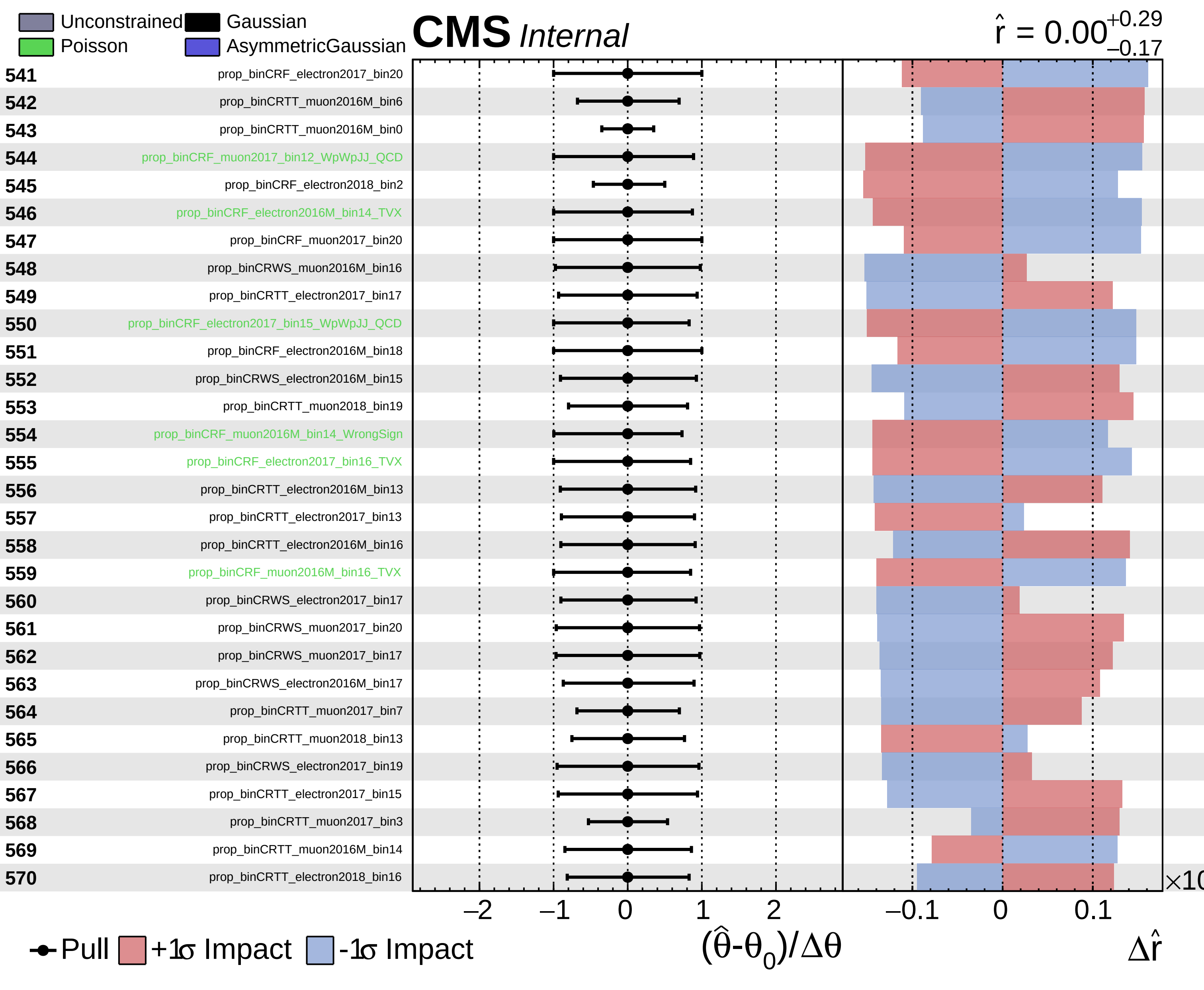
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
451 prop_binCRWS_muon2016M_bin0
452 prop_binSR_electron2017_bin19_WrongSign
453 prop_binCRTT_muon2017_bin12
454 prop_binCRTT_electron2018_bin11
455 prop_binCRWS_muon2016M_bin15
456 tau_vsjet_2016M
457 prop_binCRTT_muon2016M_bin3
458 prop_binCRF_electron2017_bin1
459 prop_binCRTT_electron2016M_bin4
460 prop_binSR_muon2016M_bin1
461 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
462 prop_binCRTT_muon2016M_bin12
463 prop_binCRTT_electron2017_bin9
464 prop_binSR_electron2017_bin11_TTTo2L2Nu
465 prop_binCRF_electron2016M_bin16
466 prop_binCRTT_electron2016M_bin2
467 prop_binCRTT_muon2016M_bin13
468 prop_binCRTT_electron2018_bin17
469 prop_binCRF_electron2017_bin18
470 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
471 prop_binCRTT_muon2018_bin11
472 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
473 prop_binCRTT_muon2016M_bin2
474 prop_binCRTT_electron2018_bin7
475 prop_binCRF_electron2018_bin1
476 prop_binCRF_muon2017_bin12_TVX
477 prop_binCRF_muon2018_bin16_TVX
478 prop_binCRWS_muon2017_bin16
479 prop_binCRWS_muon2016M_bin8
480 prop_binCRWS_muon2017_bin11_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



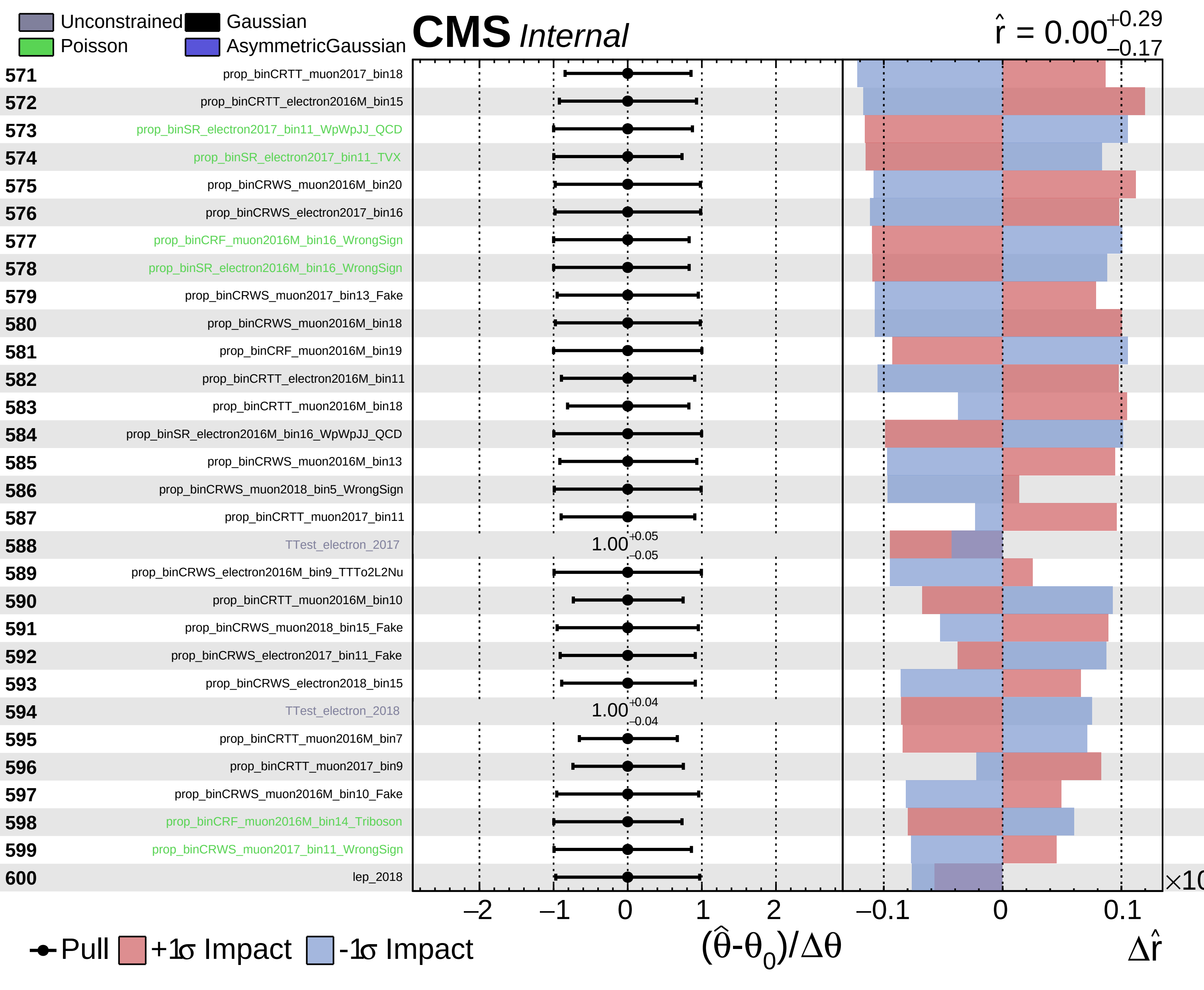
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
481 prop_binSR_electron2017_bin19_Triboson
482 prop_binCRF_electron2017_bin19
483 prop_binCRTT_muon2018_bin16
484 prop_binCRF_electron2017_bin16_WrongSign
485 prop_binCRTT_muon2016M_bin5
486 prop_binCRTT_electron2017_bin16
487 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
488 prop_binCRF_electron2016M_bin20
489 prop_binCRTT_electron2016M_bin20
490 prop_binCRF_muon2018_bin16_WrongSign
491 prop_binCRTT_muon2017_bin16
492 prop_binCRTT_muon2016M_bin1
493 prop_binCRTT_electron2017_bin14
494 prop_binCRTT_electron2017_bin8
495 prop_binCRWS_electron2018_bin18
496 prop_binCRWS_muon2016M_bin19
497 prop_binCRF_muon2017_bin18
498 prop_binCRF_muon2016M_bin14_WpWpJJ_QCD
499 prop_binCRF_muon2018_bin20
500 prop_binCRTT_electron2017_bin12
501 prop_binCRTT_muon2016M_bin8
502 prop_binCRTT_muon2016M_bin4
503 prop_binCRTT_electron2017_bin7
504 prop_binCRTT_electron2016M_bin8
505 prop_binCRWS_electron2017_bin14_Fake
506 prop_binSR_electron2017_bin14_Triboson
507 prop_binCRTT_electron2018_bin19
508 prop_binCRTT_muon2017_bin13
509 prop_binSR_electron2016M_bin0
510 prop_binCRF_muon2017_bin12_WrongSign
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



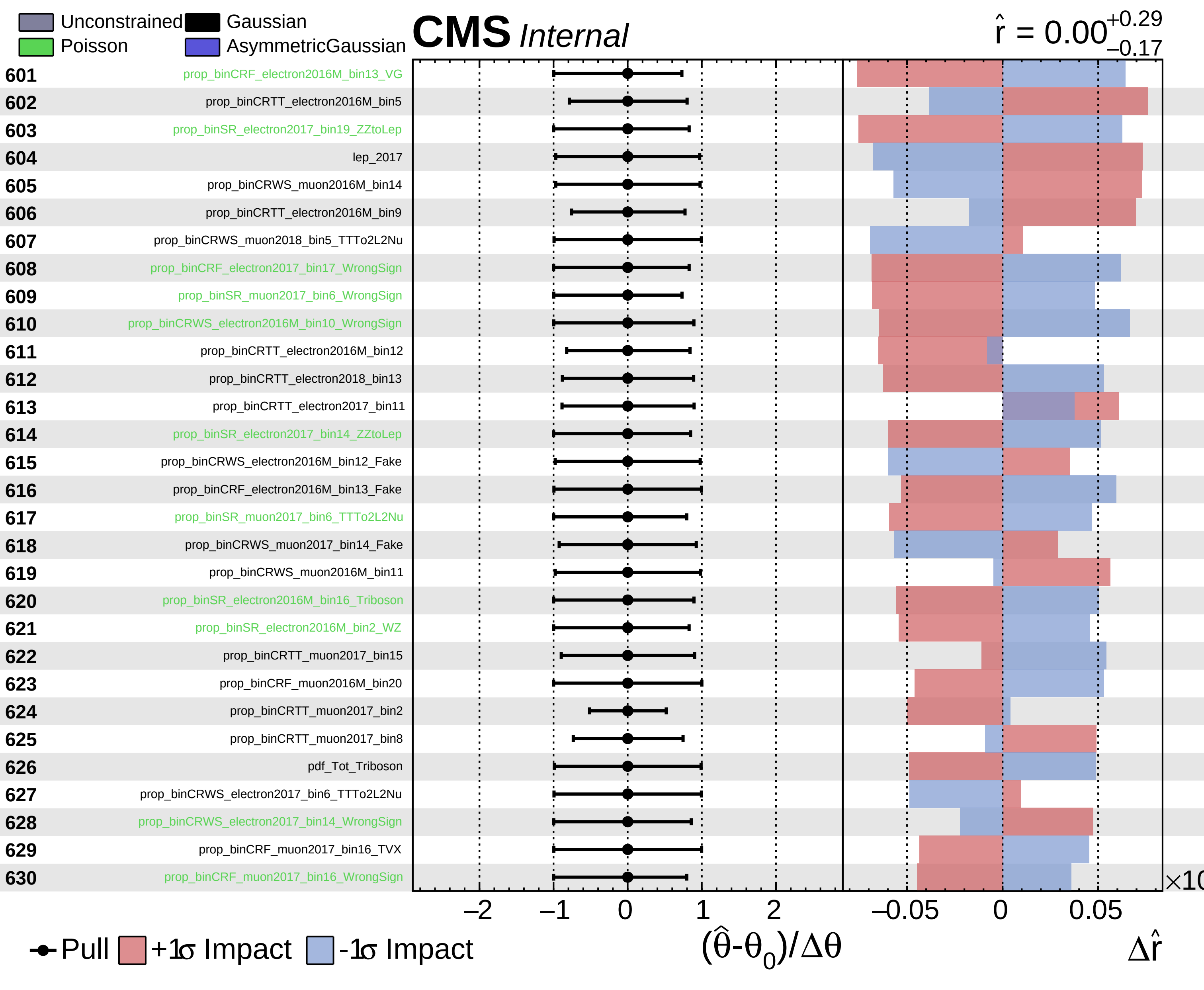
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
511 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
512 prop_binCRTT_electron2018_bin3
513 prop_binCRTT_muon2018_bin12
514 prop_binCRWS_electron2016M_bin7
515 prop_binSR_electron2016M_bin16_TVX
516 prop_binCRTT_muon2018_bin15
517 prop_binCRF_muon2018_bin16_WpWpJJ_QCD
518 prop_binCRTT_muon2016M_bin15
519 prop_binCRWS_electron2017_bin6_Fake
520 prop_binSR_electron2017_bin12_TVX
521 prop_binCRTT_electron2018_bin5
522 prop_binCRTT_electron2016M_bin17
523 prop_binCRTT_electron2016M_bin19
524 prop_binCRTT_muon2016M_bin20
525 prop_binCRTT_electron2017_bin19
526 prop_binCRWS_electron2017_bin18
527 prop_binCRTT_electron2016M_bin14
528 prop_binCRTT_electron2018_bin15
529 prop_binSR_electron2017_bin0
530 prop_binCRTT_muon2016M_bin9
531 prop_binSR_electron2017_bin12_Triboson
532 prop_binCRTT_electron2017_bin6
533 prop_binCRTT_muon2017_bin19
534 prop_binCRTT_muon2016M_bin17
535 prop_binCRWS_electron2017_bin15
536 prop_binCRWS_electron2017_bin20
537 prop_binCRWS_muon2017_bin7_Fake
538 prop_binCRTT_electron2016M_bin18
539 ISR
540 puID
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

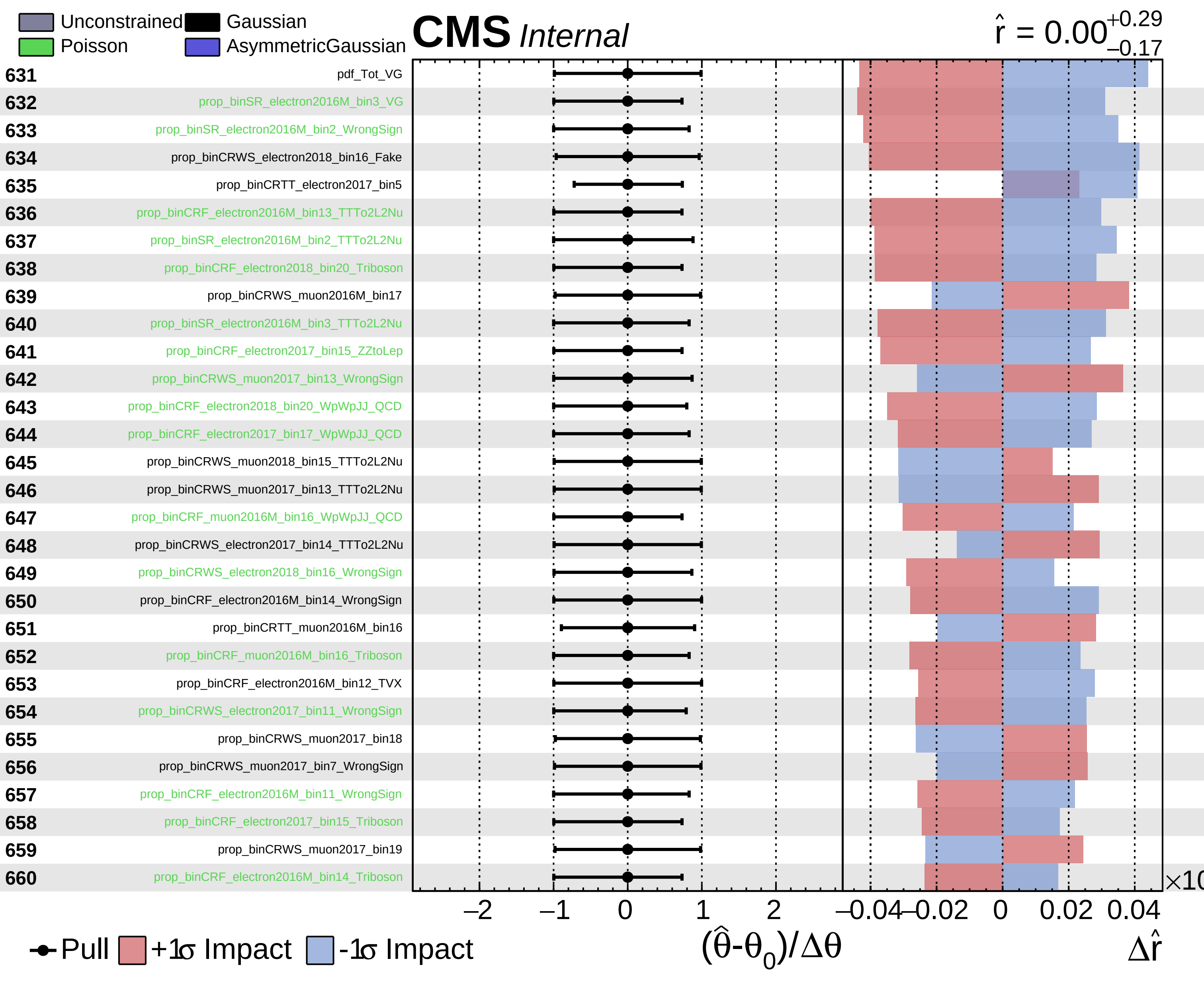


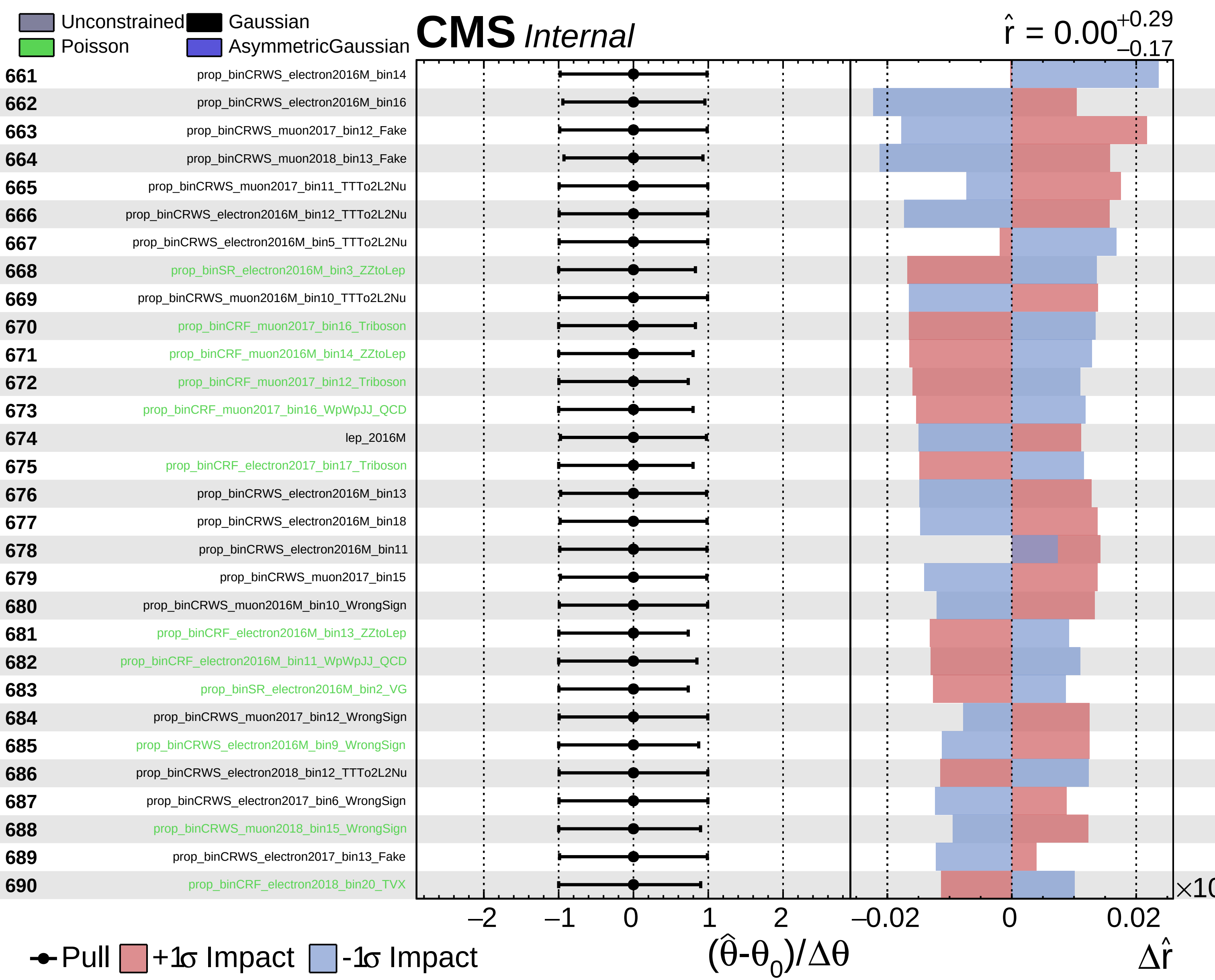
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
541 prop_binCRF_electron2017_bin20
542 prop_binCRTT_muon2016M_bin6
543 prop_binCRTT_muon2016M_bin0
544 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
545 prop_binCRF_electron2018_bin2
546 prop_binCRF_electron2016M_bin14_TVX
547 prop_binCRF_muon2017_bin20
548 prop_binCRWS_muon2016M_bin16
549 prop_binCRTT_electron2017_bin17
550 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
551 prop_binCRF_electron2016M_bin18
552 prop_binCRWS_electron2016M_bin15
553 prop_binCRTT_muon2018_bin19
554 prop_binCRF_muon2016M_bin14_WrongSign
555 prop_binCRF_electron2017_bin16_TVX
556 prop_binCRTT_electron2016M_bin13
557 prop_binCRTT_electron2017_bin13
558 prop_binCRTT_electron2016M_bin16
559 prop_binCRF_muon2016M_bin16_TVX
560 prop_binCRWS_electron2017_bin17
561 prop_binCRWS_muon2017_bin20
562 prop_binCRWS_muon2017_bin17
563 prop_binCRWS_electron2016M_bin17
564 prop_binCRTT_muon2017_bin7
565 prop_binCRTT_muon2018_bin13
566 prop_binCRWS_electron2017_bin19
567 prop_binCRTT_electron2017_bin15
568 prop_binCRTT_muon2017_bin3
569 prop_binCRTT_muon2016M_bin14
570 prop_binCRTT_electron2018_bin16
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



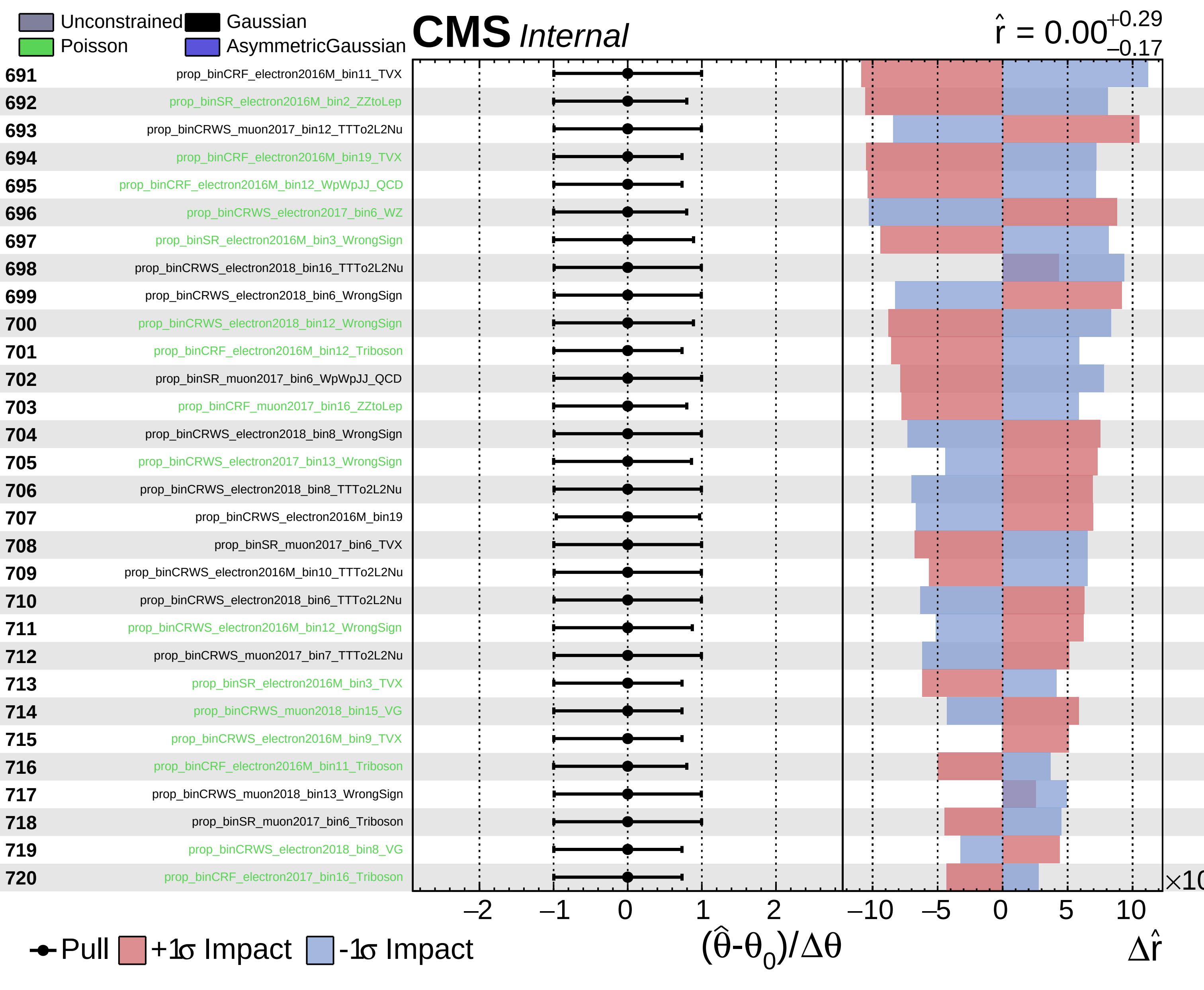
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
571 prop_binCRTT_muon2017_bin18
572 prop_binCRTT_electron2016M_bin15
573 prop_binSR_electron2017_bin11_WpWpJJ_QCD
574 prop_binSR_electron2017_bin11_TVX
575 prop_binCRWS_muon2016M_bin20
576 prop_binCRWS_electron2017_bin16
577 prop_binCRF_muon2016M_bin16_WrongSign
578 prop_binSR_electron2016M_bin16_WrongSign
579 prop_binCRWS_muon2017_bin13_Fake
580 prop_binCRWS_muon2016M_bin18
581 prop_binCRF_muon2016M_bin19
582 prop_binCRTT_electron2016M_bin11
583 prop_binCRTT_muon2016M_bin18
584 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
585 prop_binCRWS_muon2016M_bin13
586 prop_binCRWS_muon2018_bin5_WrongSign
587 prop_binCRTT_muon2017_bin11
588 TTest_electron_2017
1.00 +0.05 −0.05
589 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
590 prop_binCRTT_muon2016M_bin10
591 prop_binCRWS_muon2018_bin15_Fake
592 prop_binCRWS_electron2017_bin11_Fake
593 prop_binCRWS_electron2018_bin15
594 TTest_electron_2018
1.00 +0.04 −0.04
595 prop_binCRTT_muon2016M_bin7
596 prop_binCRTT_muon2017_bin9
597 prop_binCRWS_muon2016M_bin10_Fake
598 prop_binCRF_muon2016M_bin14_Triboson
599 prop_binCRWS_muon2017_bin11_WrongSign
600 lep_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

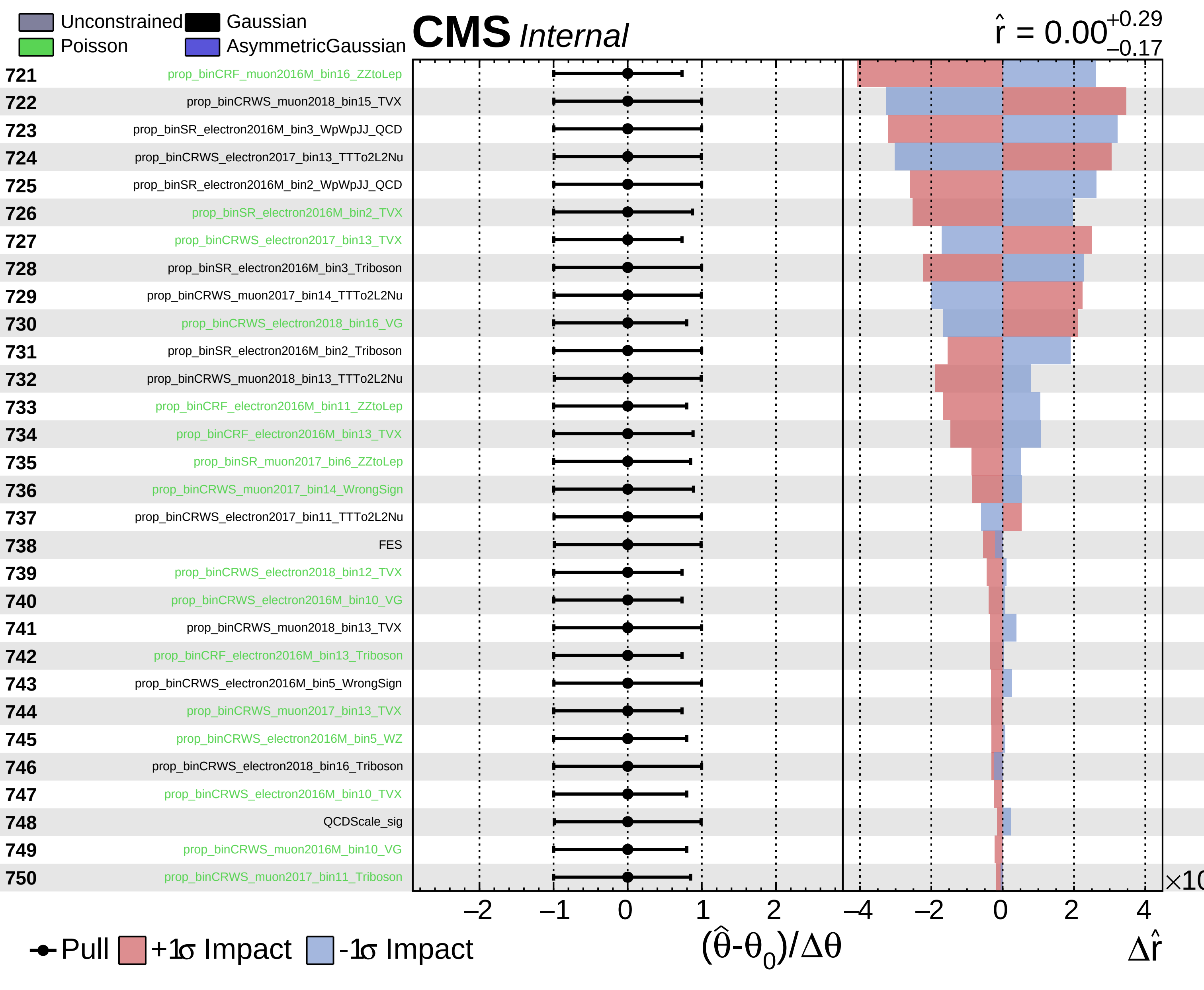

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
601 prop_binCRF_electron2016M_bin13_VG
602 prop_binCRTT_electron2016M_bin5
603 prop_binSR_electron2017_bin19_ZZtoLep
604 lep_2017
605 prop_binCRWS_muon2016M_bin14
606 prop_binCRTT_electron2016M_bin9
607 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
608 prop_binCRF_electron2017_bin17_WrongSign
609 prop_binSR_muon2017_bin6_WrongSign
610 prop_binCRWS_electron2016M_bin10_WrongSign
611 prop_binCRTT_electron2016M_bin12
612 prop_binCRTT_electron2018_bin13
613 prop_binCRTT_electron2017_bin11
614 prop_binSR_electron2017_bin14_ZZtoLep
615 prop_binCRWS_electron2016M_bin12_Fake
616 prop_binCRF_electron2016M_bin13_Fake
617 prop_binSR_muon2017_bin6_TTTo2L2Nu
618 prop_binCRWS_muon2017_bin14_Fake
619 prop_binCRWS_muon2016M_bin11
620 prop_binSR_electron2016M_bin16_Triboson
621 prop_binSR_electron2016M_bin2_WZ
622 prop_binCRTT_muon2017_bin15
623 prop_binCRF_muon2016M_bin20
624 prop_binCRTT_muon2017_bin2
625 prop_binCRTT_muon2017_bin8
626 pdf_Tot_Triboson
627 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
628 prop_binCRWS_electron2017_bin14_WrongSign
629 prop_binCRF_muon2017_bin16_TVX
630 prop_binCRF_muon2017_bin16_WrongSign
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

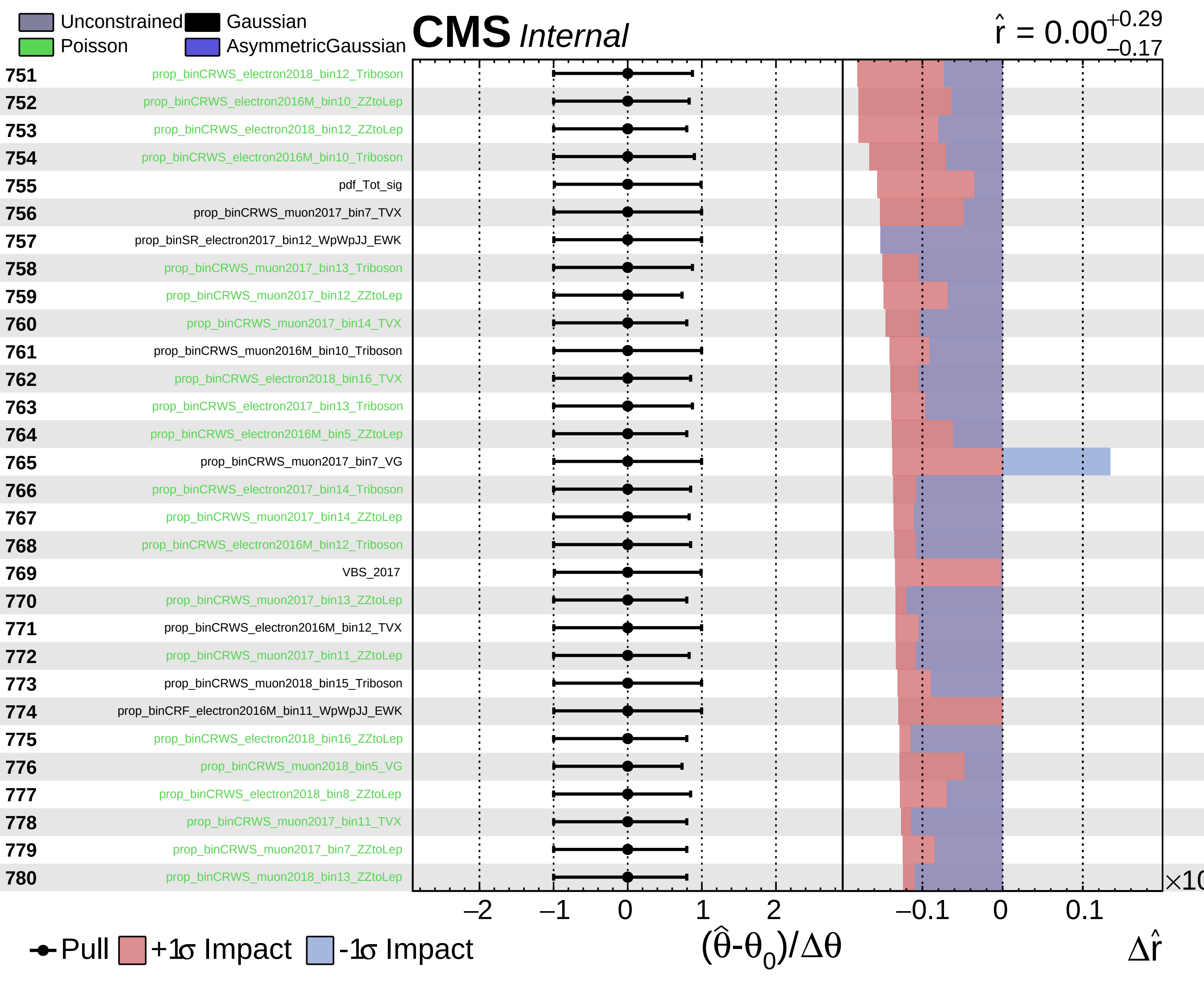

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
631 pdf_Tot_VG
632 prop_binSR_electron2016M_bin3_VG
633 prop_binSR_electron2016M_bin2_WrongSign
634 prop_binCRWS_electron2018_bin16_Fake
635 prop_binCRTT_electron2017_bin5
636 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
637 prop_binSR_electron2016M_bin2_TTTo2L2Nu
638 prop_binCRF_electron2018_bin20_Triboson
639 prop_binCRWS_muon2016M_bin17
640 prop_binSR_electron2016M_bin3_TTTo2L2Nu
641 prop_binCRF_electron2017_bin15_ZZtoLep
642 prop_binCRWS_muon2017_bin13_WrongSign
643 prop_binCRF_electron2018_bin20_WpWpJJ_QCD
644 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
645 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
646 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
647 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
648 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
649 prop_binCRWS_electron2018_bin16_WrongSign
650 prop_binCRF_electron2016M_bin14_WrongSign
651 prop_binCRTT_muon2016M_bin16
652 prop_binCRF_muon2016M_bin16_Triboson
653 prop_binCRF_electron2016M_bin12_TVX
654 prop_binCRWS_electron2017_bin11_WrongSign
655 prop_binCRWS_muon2017_bin18
656 prop_binCRWS_muon2017_bin7_WrongSign
657 prop_binCRF_electron2016M_bin11_WrongSign
658 prop_binCRF_electron2017_bin15_Triboson
659 prop_binCRWS_muon2017_bin19
660 prop_binCRF_electron2016M_bin14_Triboson
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



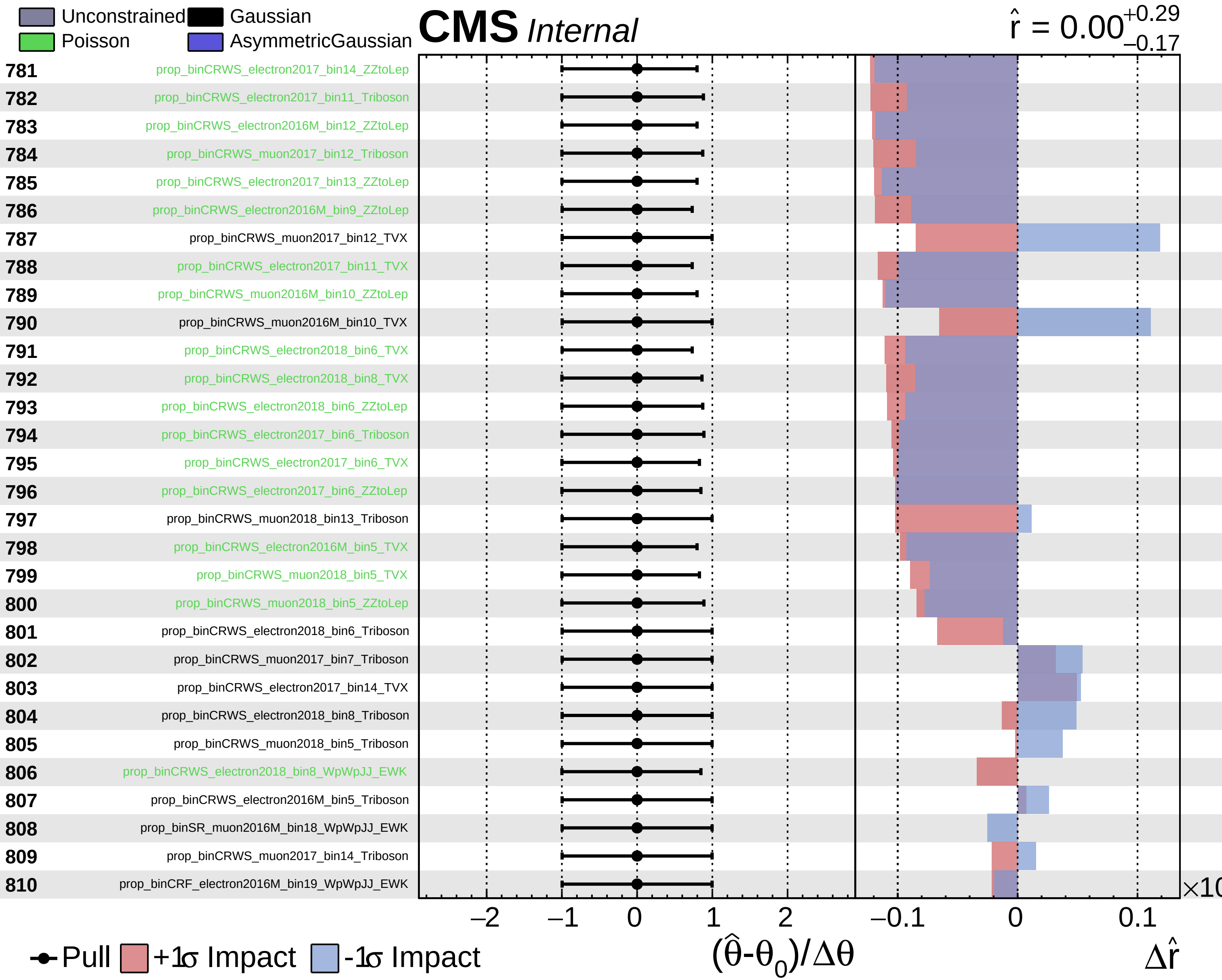
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.29 −0.17
661 prop_binCRWS_electron2016M_bin14
662 prop_binCRWS_electron2016M_bin16
663 prop_binCRWS_muon2017_bin12_Fake
664 prop_binCRWS_muon2018_bin13_Fake
665 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
666 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
667 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
668 prop_binSR_electron2016M_bin3_ZZtoLep
669 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
670 prop_binCRF_muon2017_bin16_Triboson
671 prop_binCRF_muon2016M_bin14_ZZtoLep
672 prop_binCRF_muon2017_bin12_Triboson
673 prop_binCRF_muon2017_bin16_WpWpJJ_QCD
674 lep_2016M
675 prop_binCRF_electron2017_bin17_Triboson
676 prop_binCRWS_electron2016M_bin13
677 prop_binCRWS_electron2016M_bin18
678 prop_binCRWS_electron2016M_bin11
679 prop_binCRWS_muon2017_bin15
680 prop_binCRWS_muon2016M_bin10_WrongSign
681 prop_binCRF_electron2016M_bin13_ZZtoLep
682 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
683 prop_binSR_electron2016M_bin2_VG
684 prop_binCRWS_muon2017_bin12_WrongSign
685 prop_binCRWS_electron2016M_bin9_WrongSign
686 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
687 prop_binCRWS_electron2017_bin6_WrongSign
688 prop_binCRWS_muon2018_bin15_WrongSign
689 prop_binCRWS_electron2017_bin13_Fake
690 prop_binCRF_electron2018_bin20_TVX
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
691 prop_binCRF_electron2016M_bin11_TVX
692 prop_binSR_electron2016M_bin2_ZZtoLep
693 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
694 prop_binCRF_electron2016M_bin19_TVX
695 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
696 prop_binCRWS_electron2017_bin6_WZ
697 prop_binSR_electron2016M_bin3_WrongSign
698 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
699 prop_binCRWS_electron2018_bin6_WrongSign
700 prop_binCRWS_electron2018_bin12_WrongSign
701 prop_binCRF_electron2016M_bin12_Triboson
702 prop_binSR_muon2017_bin6_WpWpJJ_QCD
703 prop_binCRF_muon2017_bin16_ZZtoLep
704 prop_binCRWS_electron2018_bin8_WrongSign
705 prop_binCRWS_electron2017_bin13_WrongSign
706 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
707 prop_binCRWS_electron2016M_bin19
708 prop_binSR_muon2017_bin6_TVX
709 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
710 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
711 prop_binCRWS_electron2016M_bin12_WrongSign
712 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
713 prop_binSR_electron2016M_bin3_TVX
714 prop_binCRWS_muon2018_bin15_VG
715 prop_binCRWS_electron2016M_bin9_TVX
716 prop_binCRF_electron2016M_bin11_Triboson
717 prop_binCRWS_muon2018_bin13_WrongSign
718 prop_binSR_muon2017_bin6_Triboson
719 prop_binCRWS_electron2018_bin8_VG
720 prop_binCRF_electron2017_bin16_Triboson
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

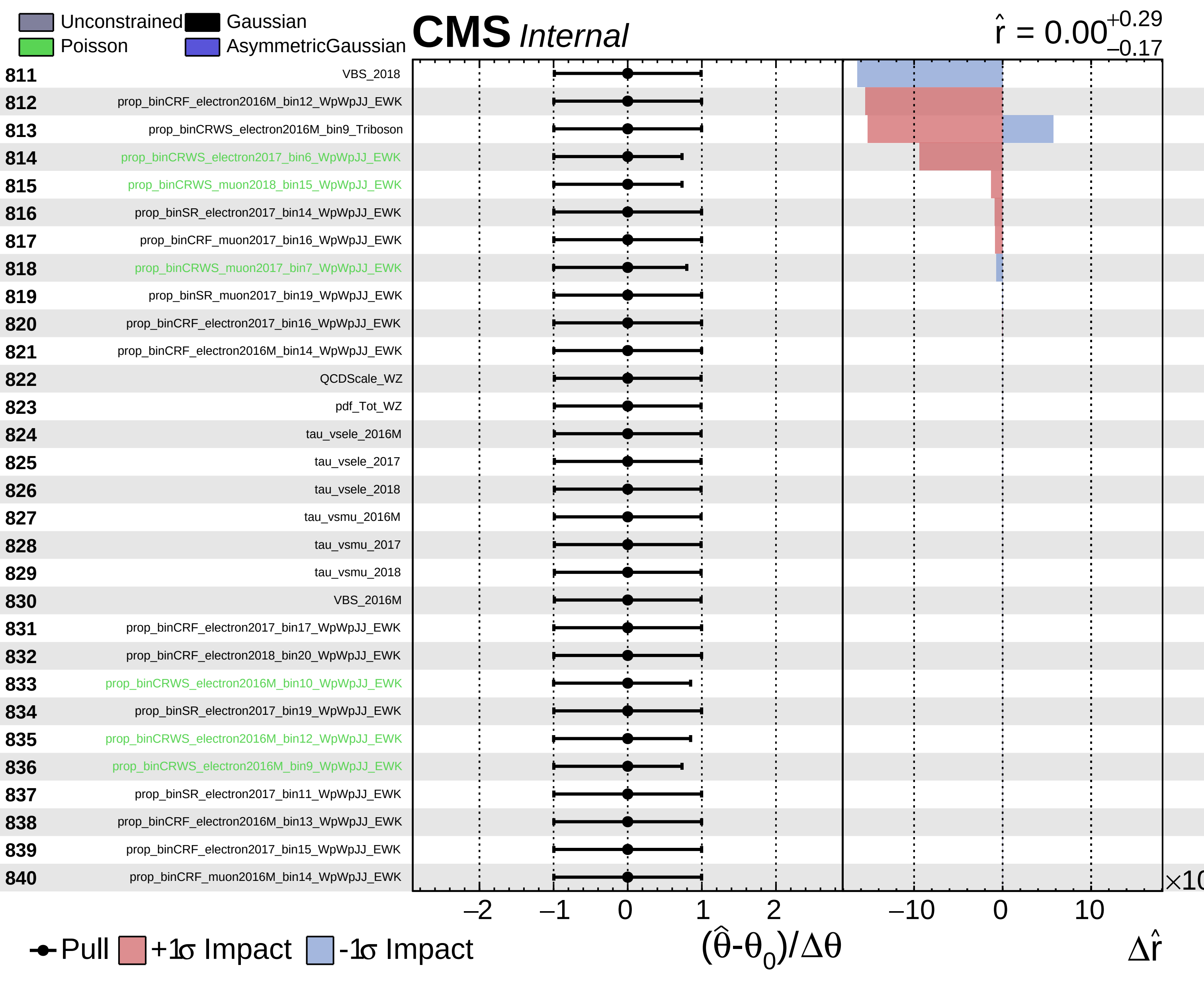

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
721 prop_binCRF_muon2016M_bin16_ZZtoLep
722 prop_binCRWS_muon2018_bin15_TVX
723 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
724 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
725 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
726 prop_binSR_electron2016M_bin2_TVX
727 prop_binCRWS_electron2017_bin13_TVX
728 prop_binSR_electron2016M_bin3_Triboson
729 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
730 prop_binCRWS_electron2018_bin16_VG
731 prop_binSR_electron2016M_bin2_Triboson
732 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
733 prop_binCRF_electron2016M_bin11_ZZtoLep
734 prop_binCRF_electron2016M_bin13_TVX
735 prop_binSR_muon2017_bin6_ZZtoLep
736 prop_binCRWS_muon2017_bin14_WrongSign
737 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
738 FES
739 prop_binCRWS_electron2018_bin12_TVX
740 prop_binCRWS_electron2016M_bin10_VG
741 prop_binCRWS_muon2018_bin13_TVX
742 prop_binCRF_electron2016M_bin13_Triboson
743 prop_binCRWS_electron2016M_bin5_WrongSign
744 prop_binCRWS_muon2017_bin13_TVX
745 prop_binCRWS_electron2016M_bin5_WZ
746 prop_binCRWS_electron2018_bin16_Triboson
747 prop_binCRWS_electron2016M_bin10_TVX
748 QCDScale_sig
749 prop_binCRWS_muon2016M_bin10_VG
750 prop_binCRWS_muon2017_bin11_Triboson
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



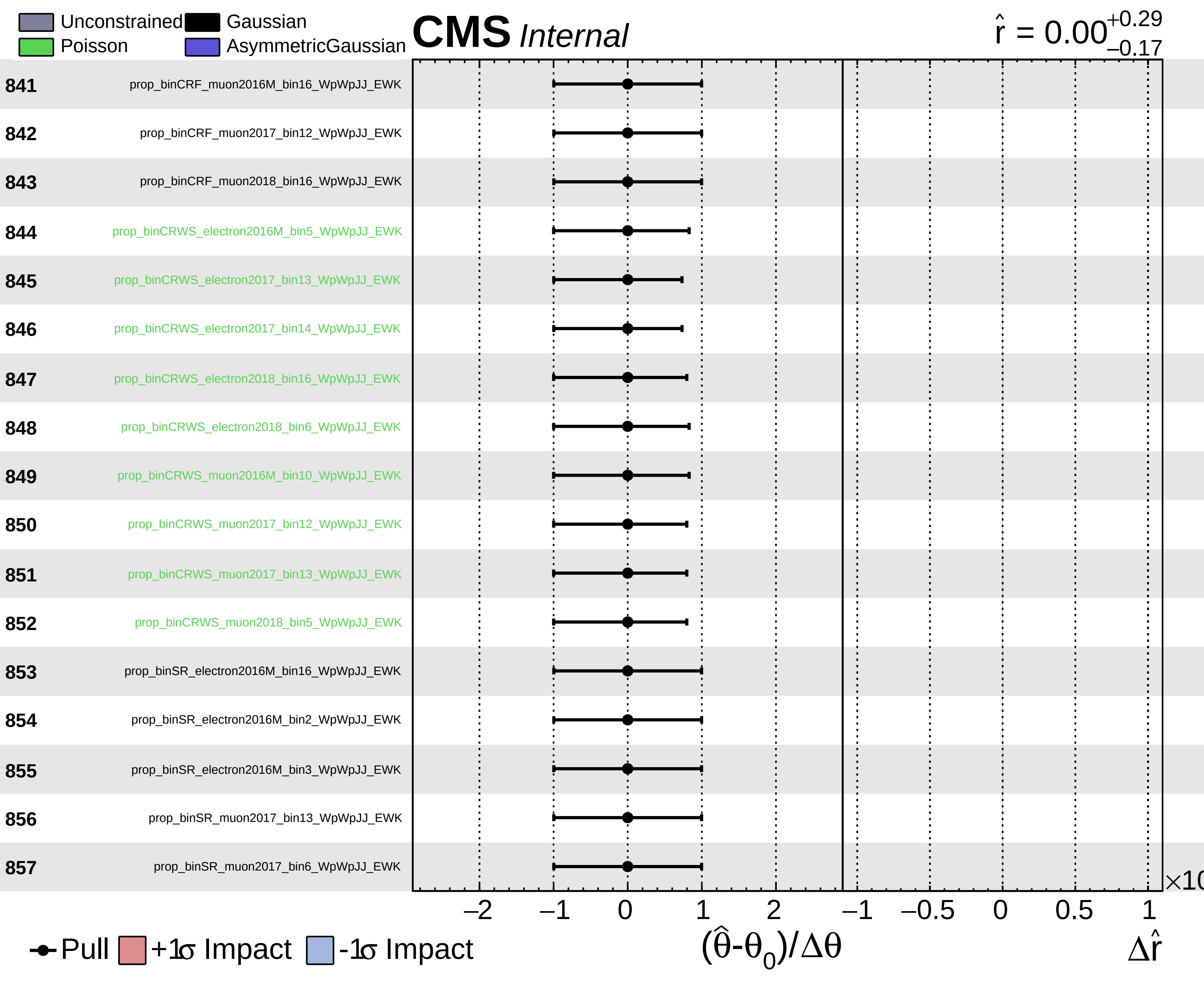
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
751 prop_binCRWS_electron2018_bin12_Triboson
752 prop_binCRWS_electron2016M_bin10_ZZtoLep
753 prop_binCRWS_electron2018_bin12_ZZtoLep
754 prop_binCRWS_electron2016M_bin10_Triboson
755 pdf_Tot_sig
756 prop_binCRWS_muon2017_bin7_TVX
757 prop_binSR_electron2017_bin12_WpWpJJ_EWK
758 prop_binCRWS_muon2017_bin13_Triboson
759 prop_binCRWS_muon2017_bin12_ZZtoLep
760 prop_binCRWS_muon2017_bin14_TVX
761 prop_binCRWS_muon2016M_bin10_Triboson
762 prop_binCRWS_electron2018_bin16_TVX
763 prop_binCRWS_electron2017_bin13_Triboson
764 prop_binCRWS_electron2016M_bin5_ZZtoLep
765 prop_binCRWS_muon2017_bin7_VG
766 prop_binCRWS_electron2017_bin14_Triboson
767 prop_binCRWS_muon2017_bin14_ZZtoLep
768 prop_binCRWS_electron2016M_bin12_Triboson
769 VBS_2017
770 prop_binCRWS_muon2017_bin13_ZZtoLep
771 prop_binCRWS_electron2016M_bin12_TVX
772 prop_binCRWS_muon2017_bin11_ZZtoLep
773 prop_binCRWS_muon2018_bin15_Triboson
774 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
775 prop_binCRWS_electron2018_bin16_ZZtoLep
776 prop_binCRWS_muon2018_bin5_VG
777 prop_binCRWS_electron2018_bin8_ZZtoLep
778 prop_binCRWS_muon2017_bin11_TVX
779 prop_binCRWS_muon2017_bin7_ZZtoLep
780 prop_binCRWS_muon2018_bin13_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
781 prop_binCRWS_electron2017_bin14_ZZtoLep
782 prop_binCRWS_electron2017_bin11_Triboson
783 prop_binCRWS_electron2016M_bin12_ZZtoLep
784 prop_binCRWS_muon2017_bin12_Triboson
785 prop_binCRWS_electron2017_bin13_ZZtoLep
786 prop_binCRWS_electron2016M_bin9_ZZtoLep
787 prop_binCRWS_muon2017_bin12_TVX
788 prop_binCRWS_electron2017_bin11_TVX
789 prop_binCRWS_muon2016M_bin10_ZZtoLep
790 prop_binCRWS_muon2016M_bin10_TVX
791 prop_binCRWS_electron2018_bin6_TVX
792 prop_binCRWS_electron2018_bin8_TVX
793 prop_binCRWS_electron2018_bin6_ZZtoLep
794 prop_binCRWS_electron2017_bin6_Triboson
795 prop_binCRWS_electron2017_bin6_TVX
796 prop_binCRWS_electron2017_bin6_ZZtoLep
797 prop_binCRWS_muon2018_bin13_Triboson
798 prop_binCRWS_electron2016M_bin5_TVX
799 prop_binCRWS_muon2018_bin5_TVX
800 prop_binCRWS_muon2018_bin5_ZZtoLep
801 prop_binCRWS_electron2018_bin6_Triboson
802 prop_binCRWS_muon2017_bin7_Triboson
803 prop_binCRWS_electron2017_bin14_TVX
804 prop_binCRWS_electron2018_bin8_Triboson
805 prop_binCRWS_muon2018_bin5_Triboson
806 prop_binCRWS_electron2018_bin8_WpWpJJ_EWK
807 prop_binCRWS_electron2016M_bin5_Triboson
808 prop_binSR_muon2016M_bin18_WpWpJJ_EWK
809 prop_binCRWS_muon2017_bin14_Triboson
810 prop_binCRF_electron2016M_bin19_WpWpJJ_EWK
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
811 VBS_2018
812 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
813 prop_binCRWS_electron2016M_bin9_Triboson
814 prop_binCRWS_electron2017_bin6_WpWpJJ_EWK
815 prop_binCRWS_muon2018_bin15_WpWpJJ_EWK
816 prop_binSR_electron2017_bin14_WpWpJJ_EWK
817 prop_binCRF_muon2017_bin16_WpWpJJ_EWK
818 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
819 prop_binSR_muon2017_bin19_WpWpJJ_EWK
820 prop_binCRF_electron2017_bin16_WpWpJJ_EWK
821 prop_binCRF_electron2016M_bin14_WpWpJJ_EWK
822 QCDScale_WZ
823 pdf_Tot_WZ
824 tau_vsele_2016M
825 tau_vsele_2017
826 tau_vsele_2018
827 tau_vsmu_2016M
828 tau_vsmu_2017
829 tau_vsmu_2018
830 VBS_2016M
831 prop_binCRF_electron2017_bin17_WpWpJJ_EWK
832 prop_binCRF_electron2018_bin20_WpWpJJ_EWK
833 prop_binCRWS_electron2016M_bin10_WpWpJJ_EWK
834 prop_binSR_electron2017_bin19_WpWpJJ_EWK
835 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
836 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
837 prop_binSR_electron2017_bin11_WpWpJJ_EWK
838 prop_binCRF_electron2016M_bin13_WpWpJJ_EWK
839 prop_binCRF_electron2017_bin15_WpWpJJ_EWK
840 prop_binCRF_muon2016M_bin14_WpWpJJ_EWK
−2 −1 0 1 2
−10 0 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.29}_{-0.17}$
841 prop_binCRF_muon2016M_bin16_WpWpJJ_EWK
842 prop_binCRF_muon2017_bin12_WpWpJJ_EWK
843 prop_binCRF_muon2018_bin16_WpWpJJ_EWK
844 prop_binCRWS_electron2016M_bin5_WpWpJJ_EWK
845 prop_binCRWS_electron2017_bin13_WpWpJJ_EWK
846 prop_binCRWS_electron2017_bin14_WpWpJJ_EWK
847 prop_binCRWS_electron2018_bin16_WpWpJJ_EWK
848 prop_binCRWS_electron2018_bin6_WpWpJJ_EWK
849 prop_binCRWS_muon2016M_bin10_WpWpJJ_EWK
850 prop_binCRWS_muon2017_bin12_WpWpJJ_EWK
851 prop_binCRWS_muon2017_bin13_WpWpJJ_EWK
852 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
853 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
854 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
855 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
856 prop_binSR_muon2017_bin13_WpWpJJ_EWK
857 prop_binSR_muon2017_bin6_WpWpJJ_EWK
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$